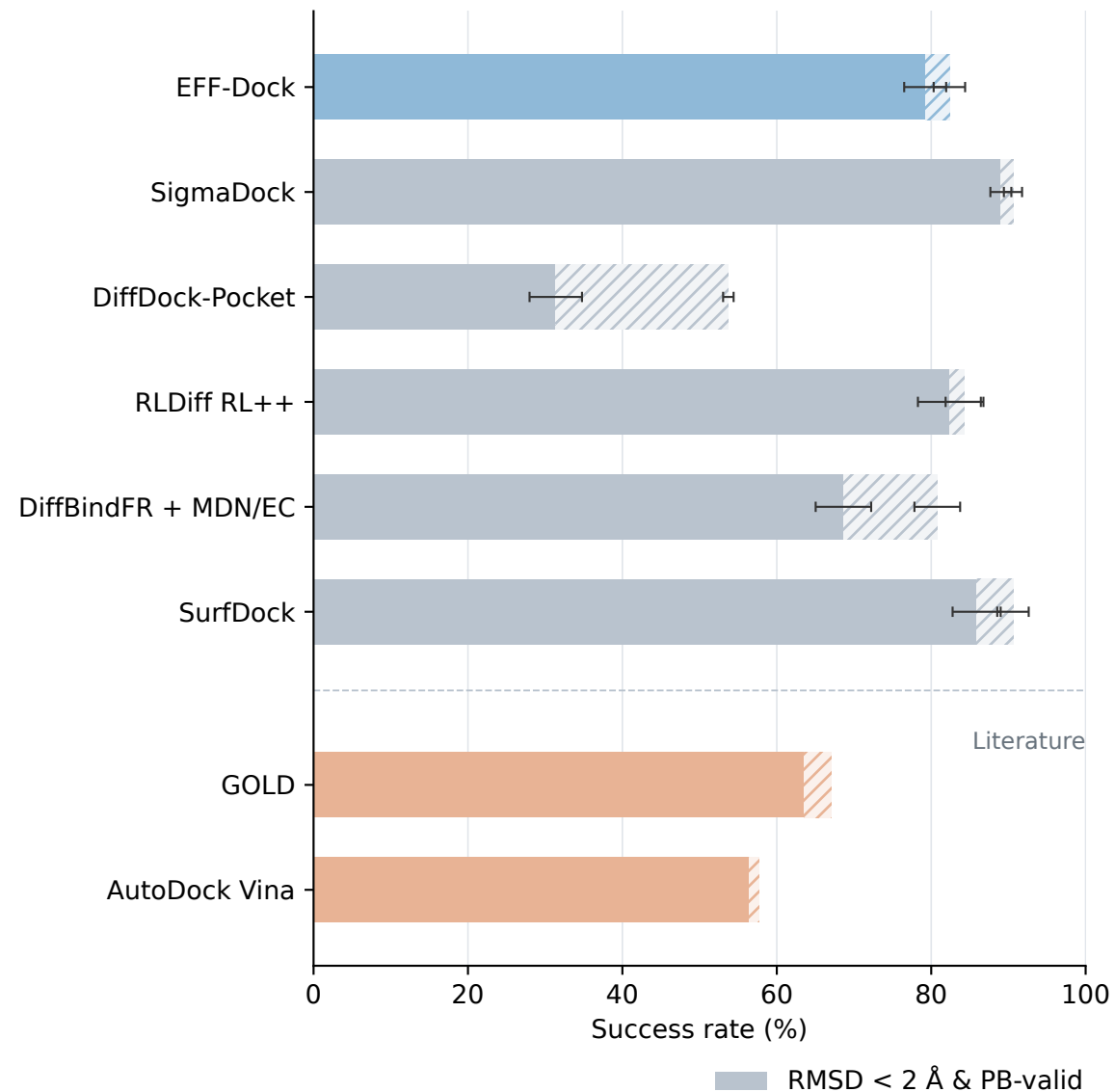
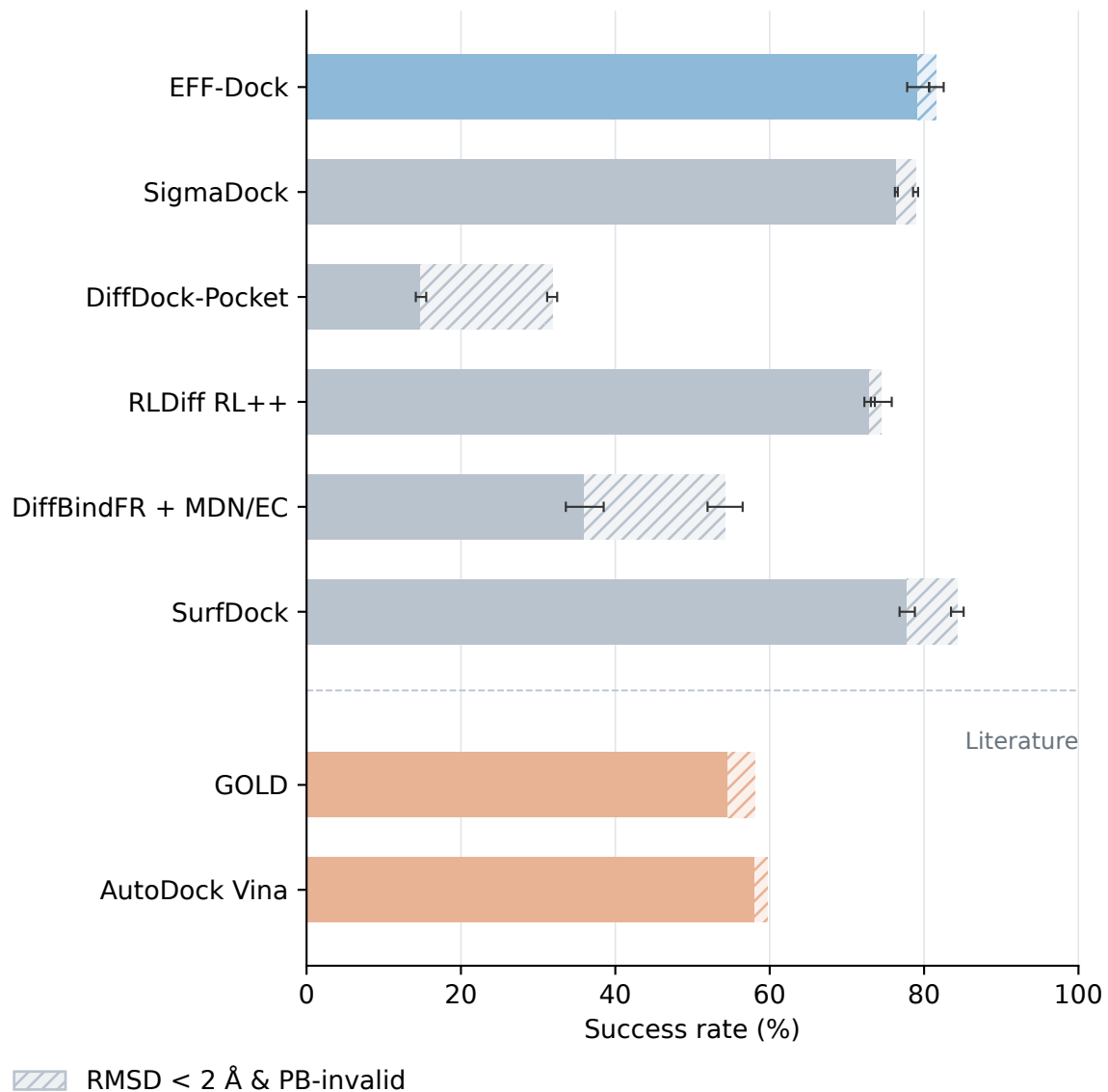
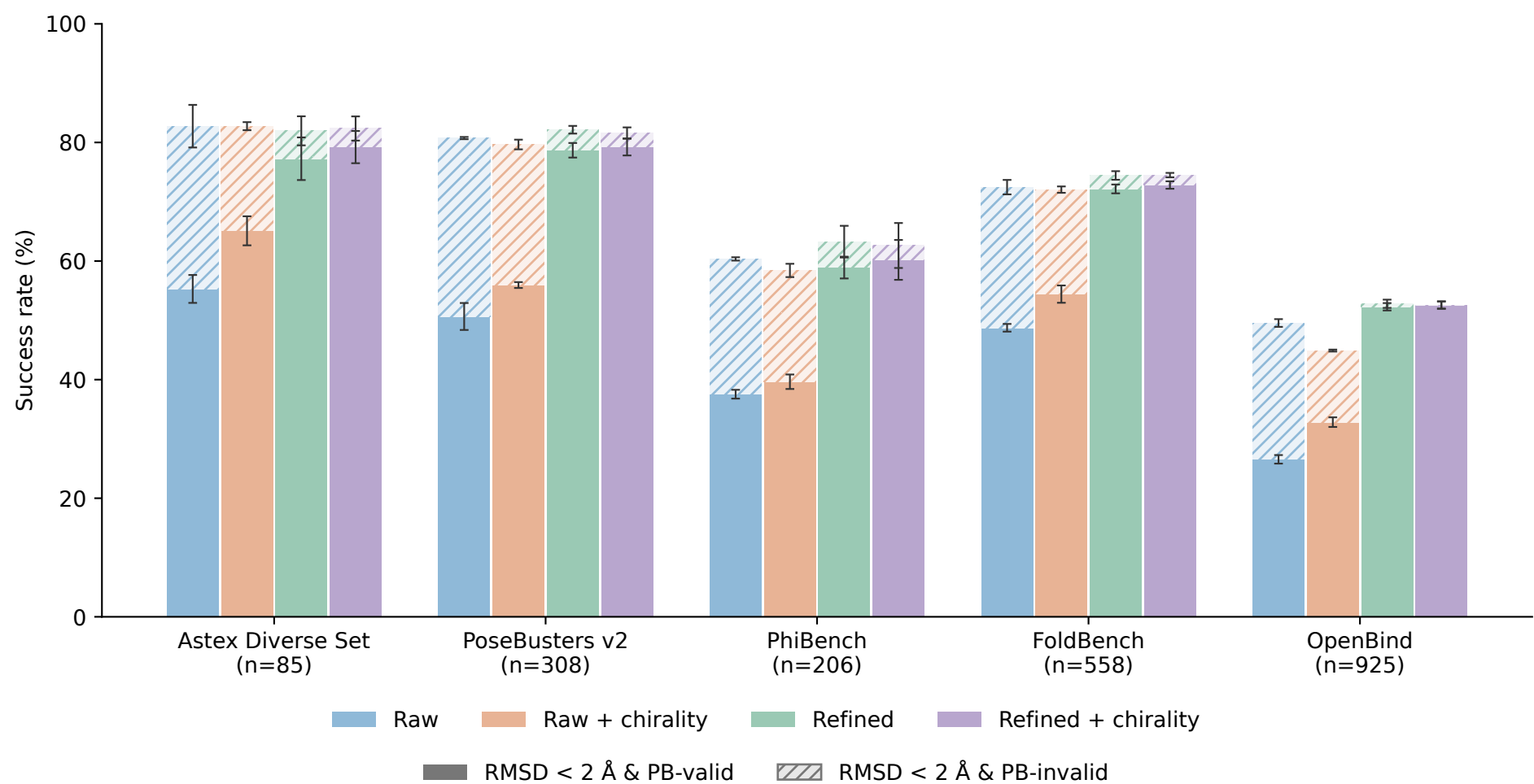
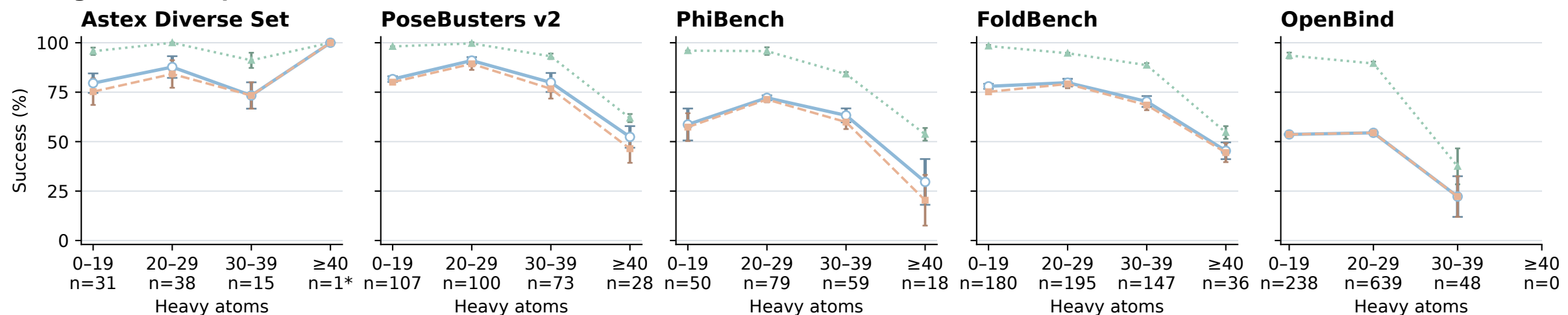


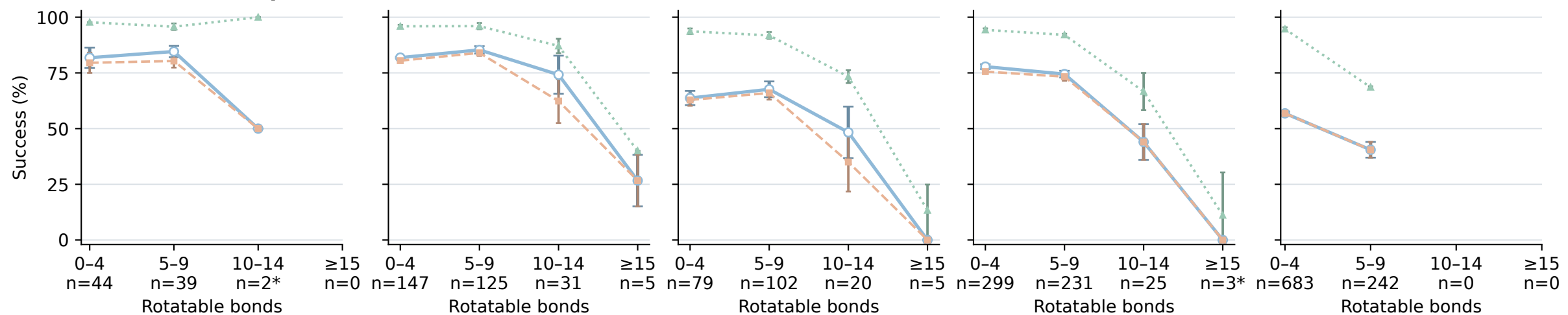
A Astex Diverse Set**B PoseBusters v2**



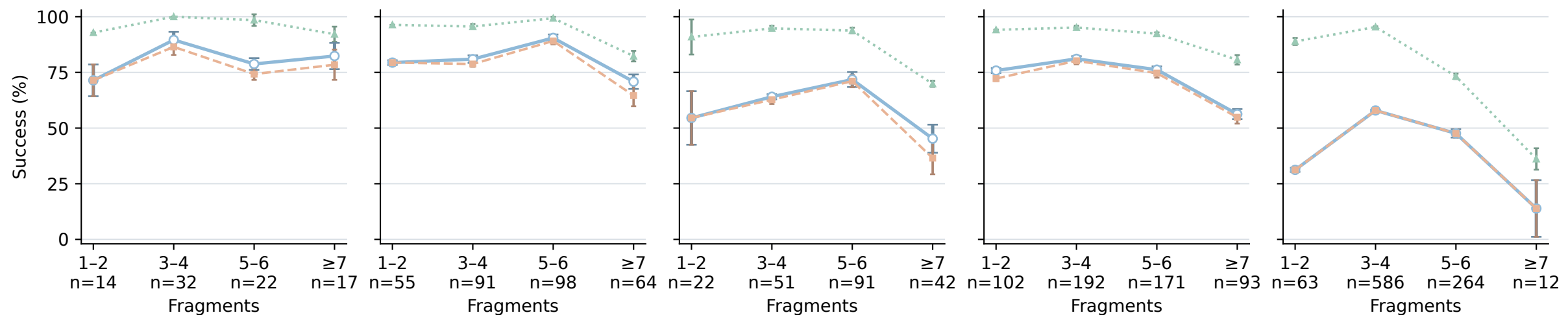
A Ligand size dependence



B Rotatable bond dependence

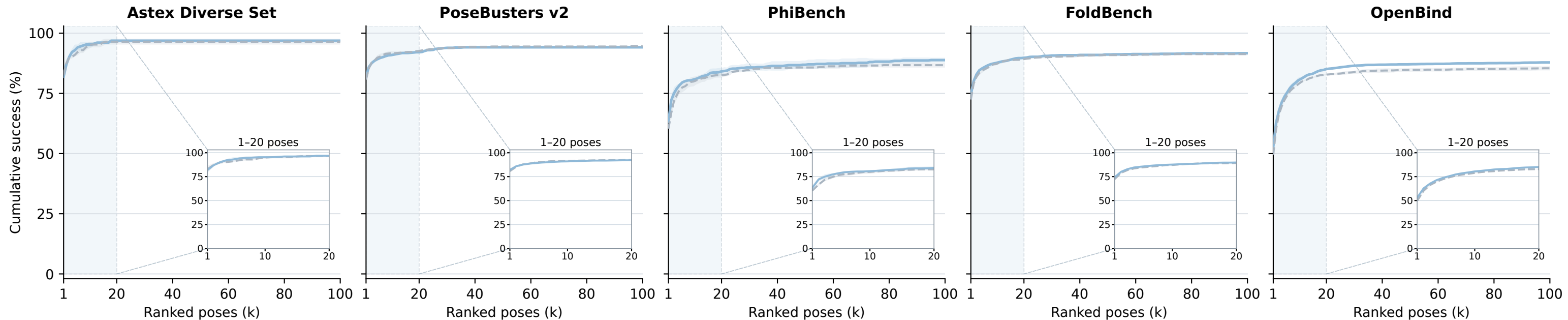


C Fragment count dependence

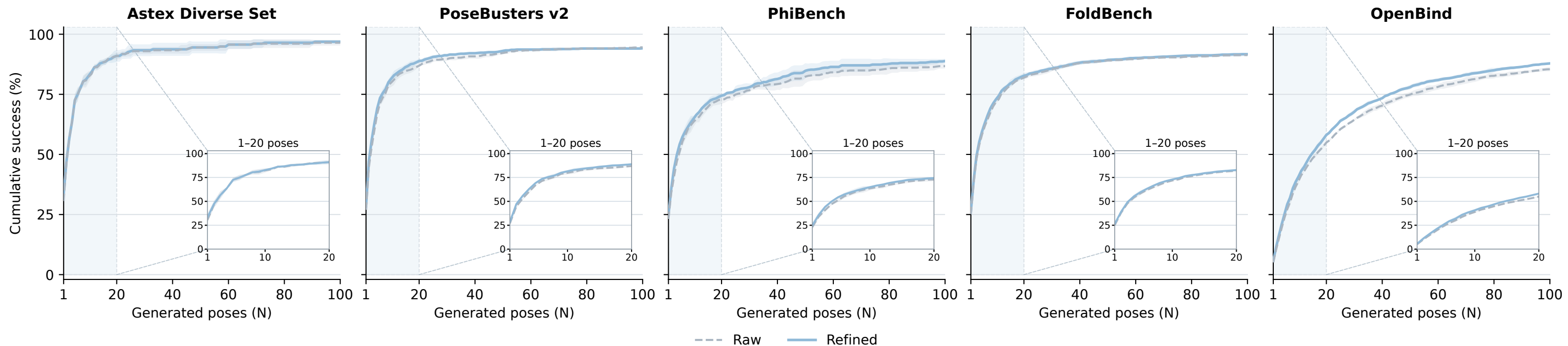


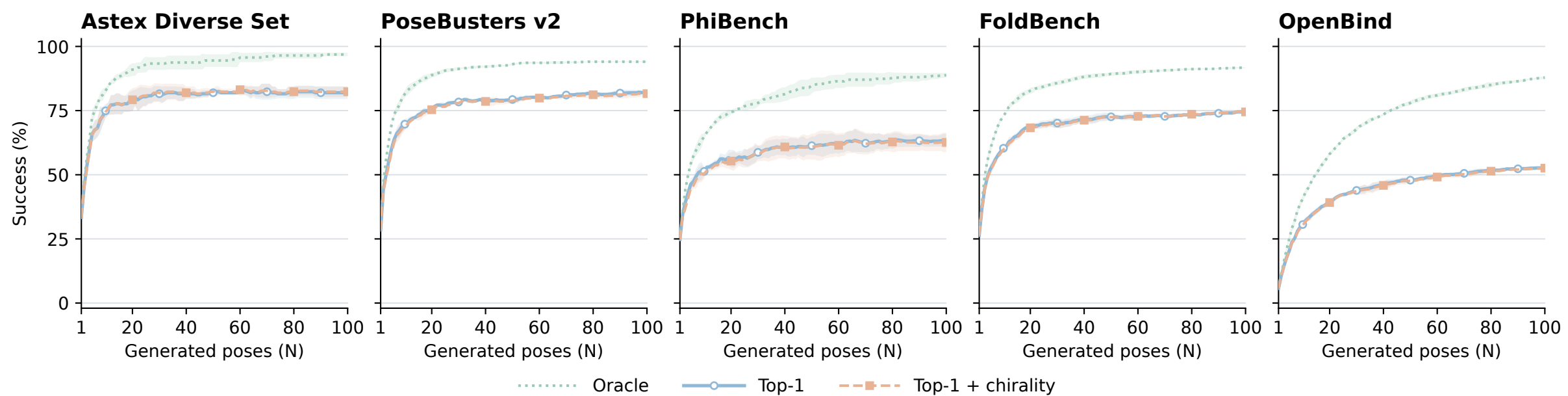
Oracle Top-1 Top-1 + PB-valid

A Confidence-ranked poses

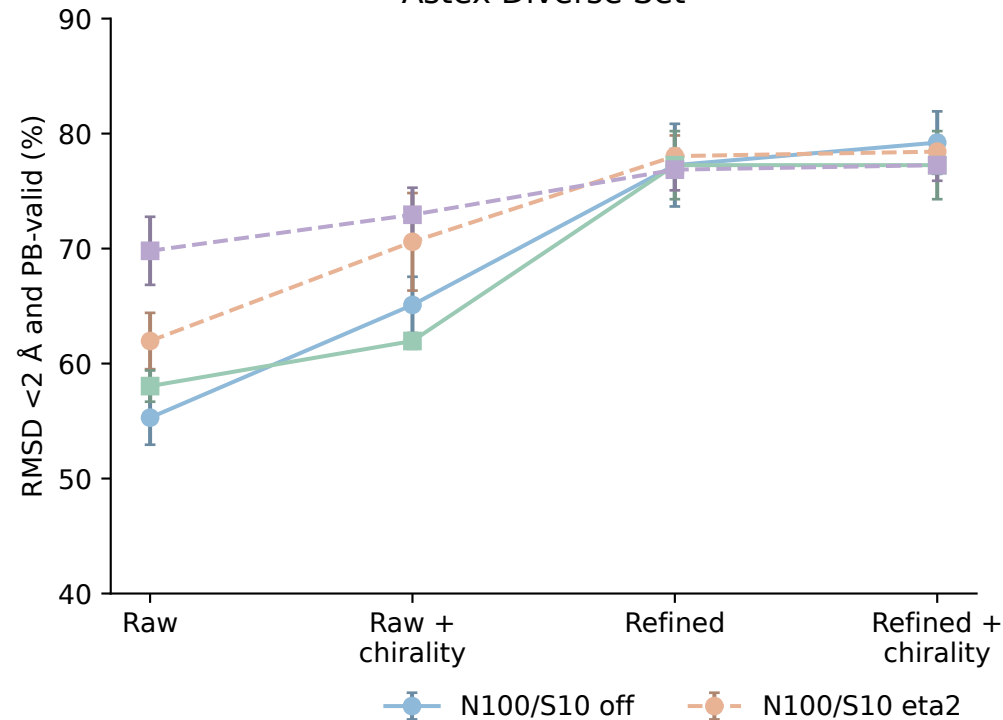


B Generation-order poses

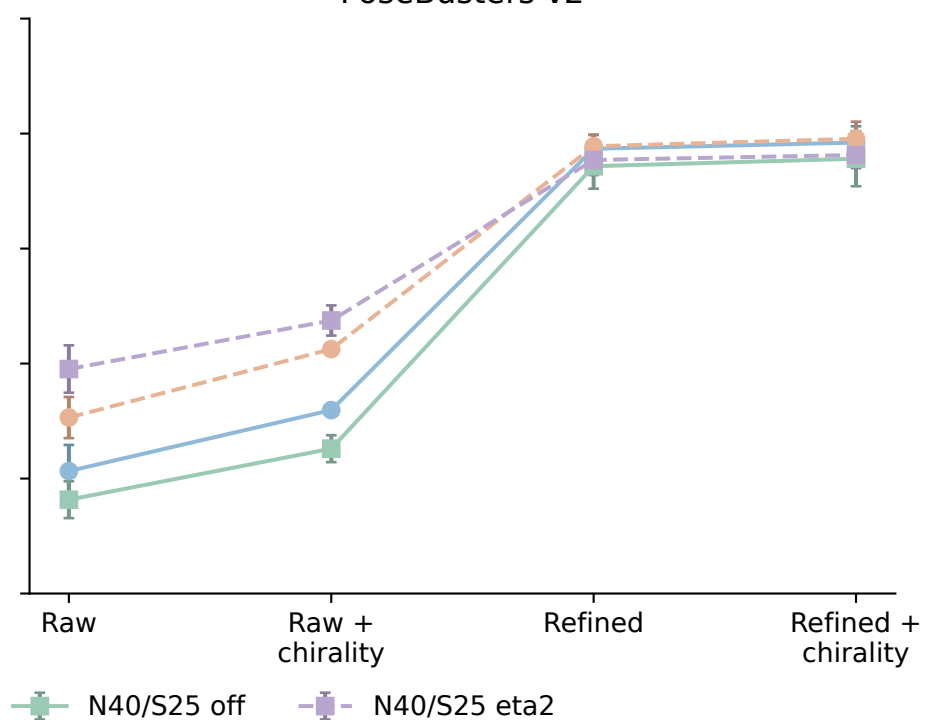


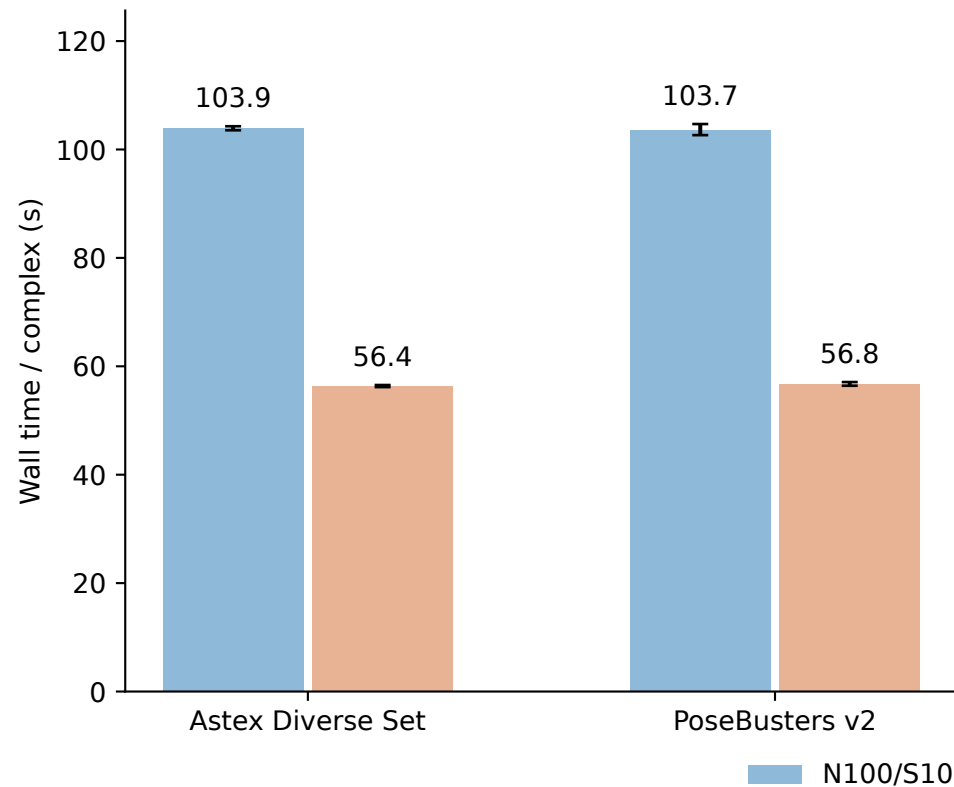
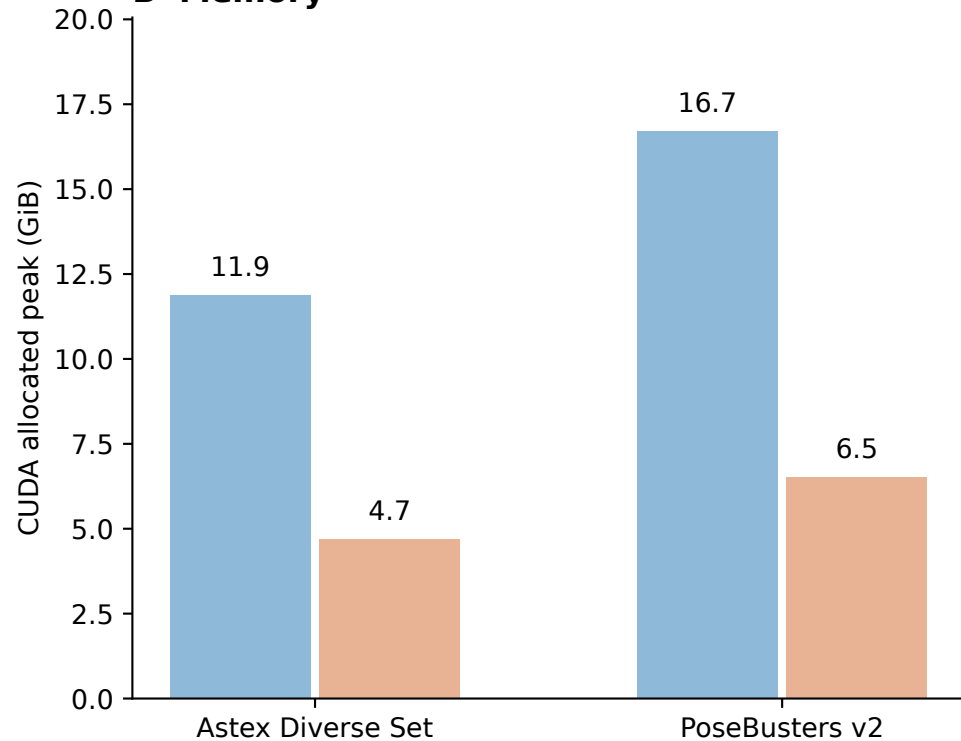


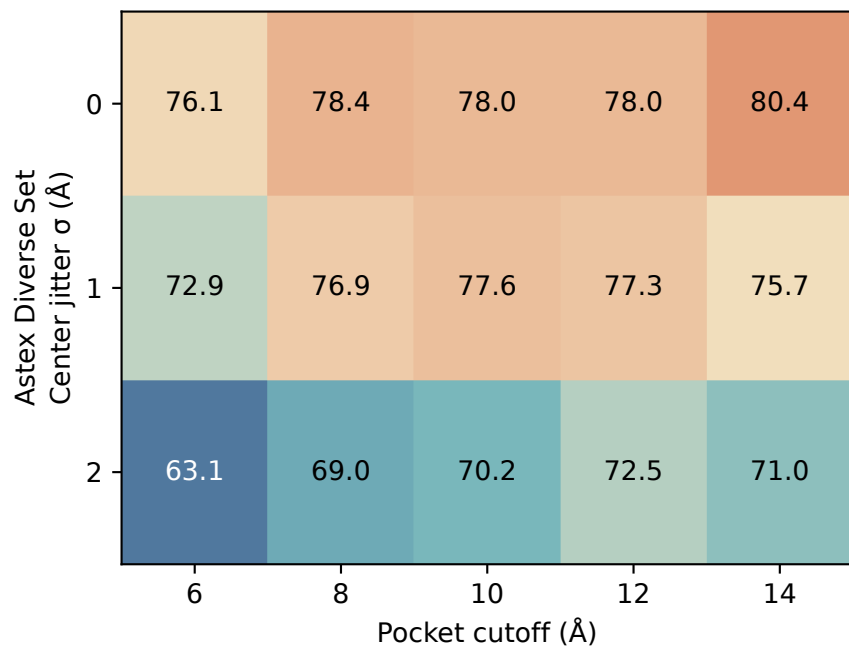
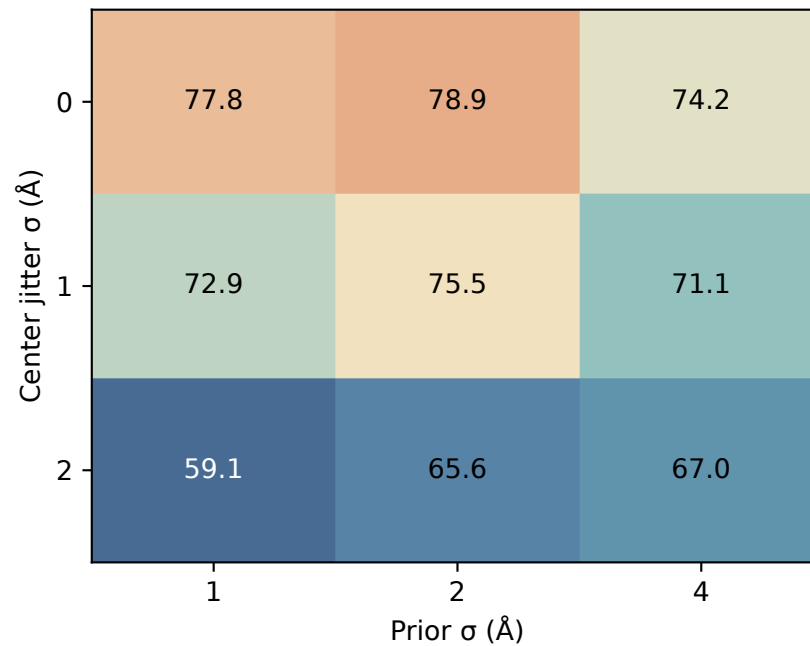
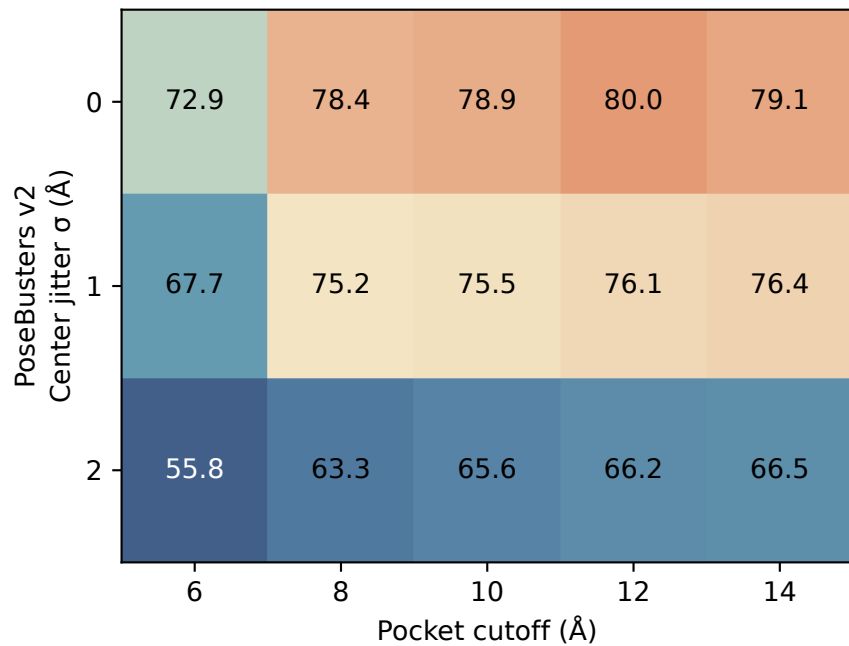
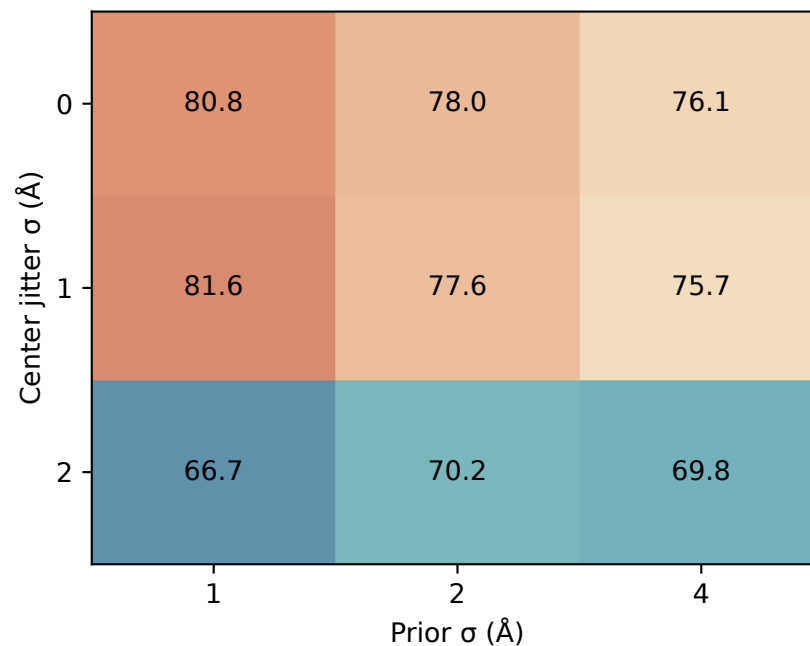
Astex Diverse Set



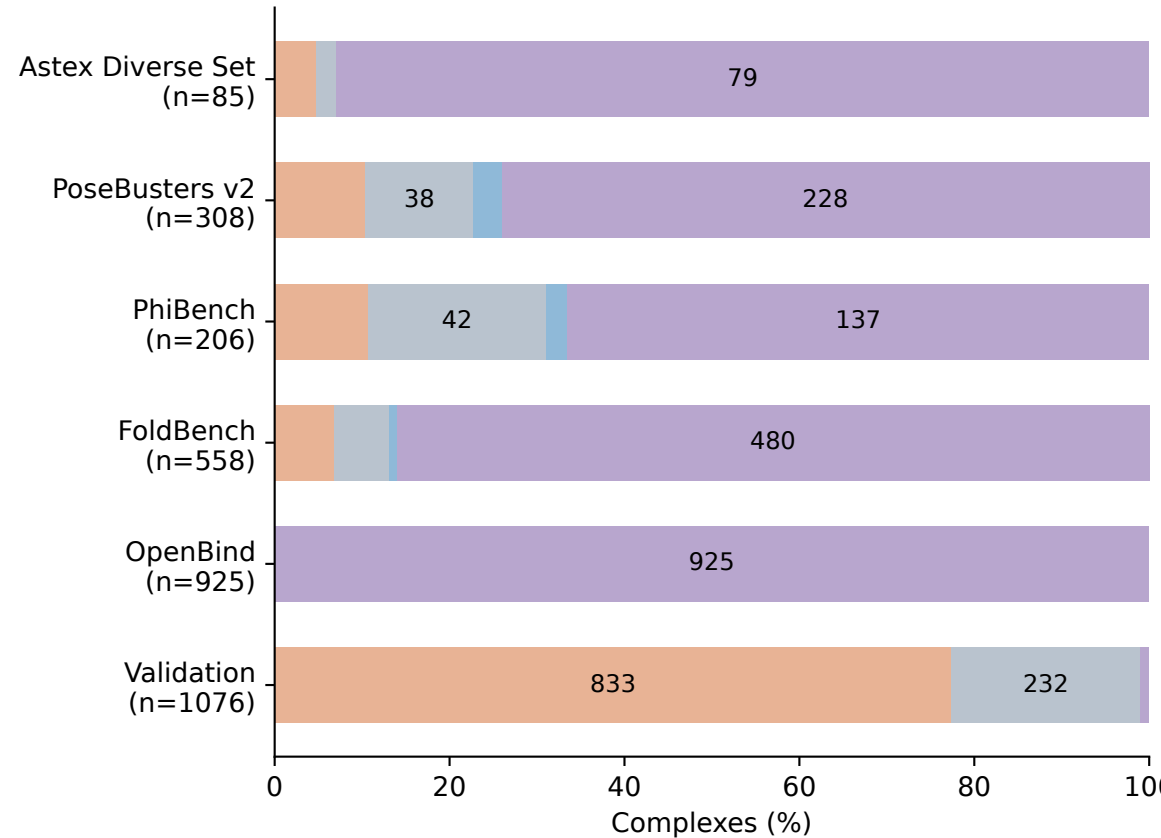
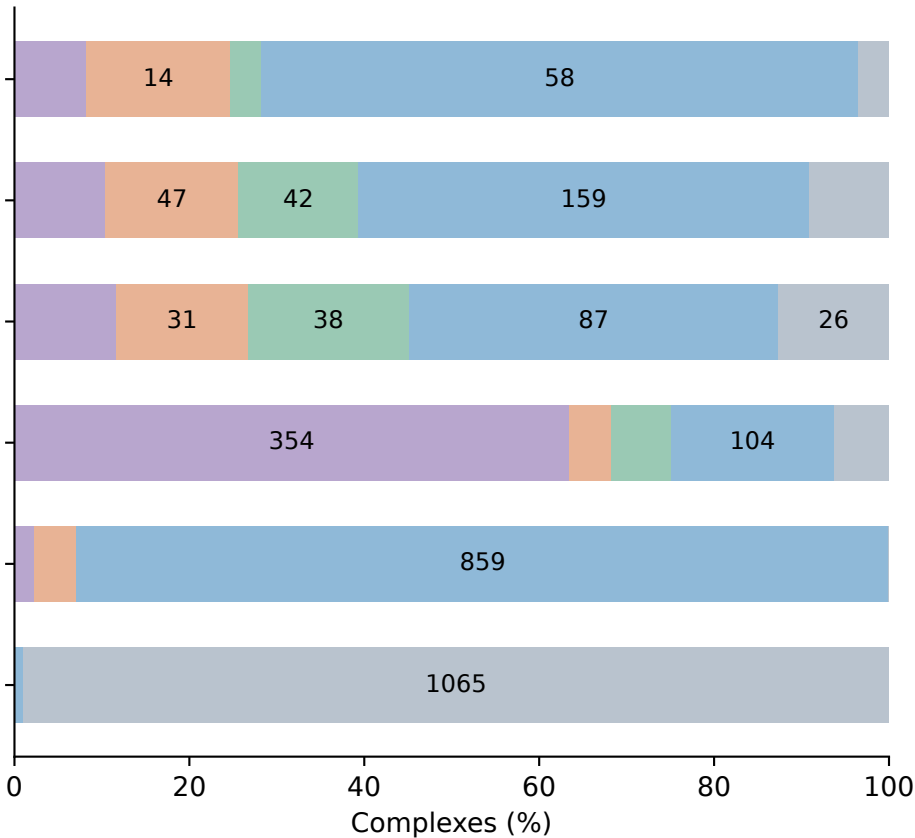
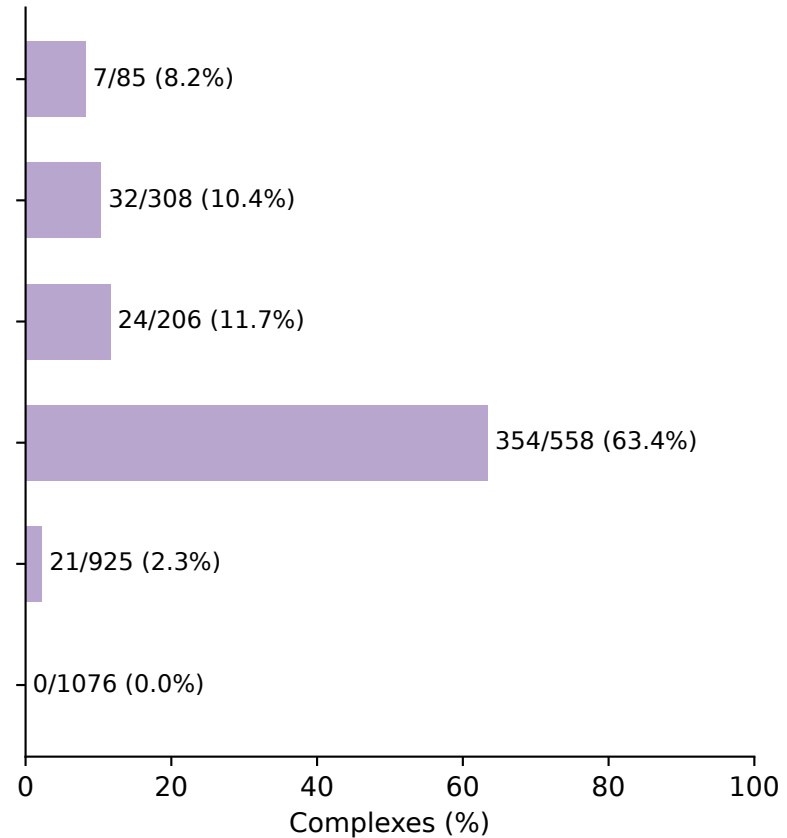
PoseBusters v2



A Runtime**B Memory**

A Pocket size**B Prior scale**

RMSD < 2 Å & PB-valid (%)

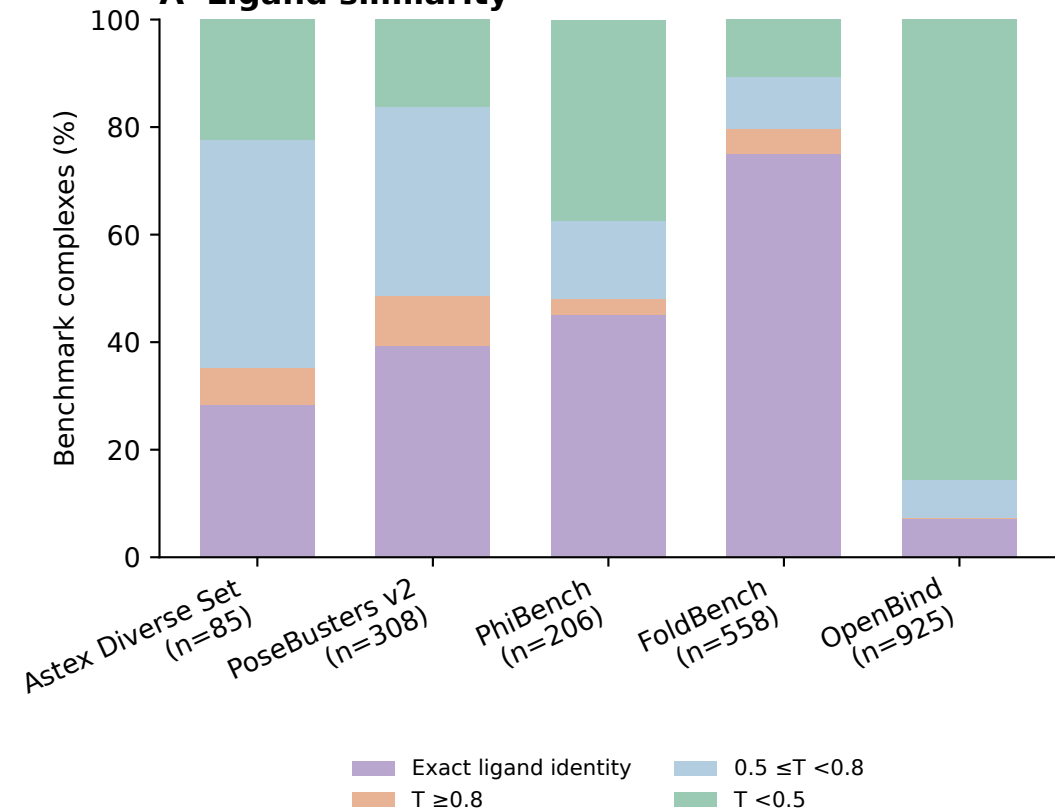
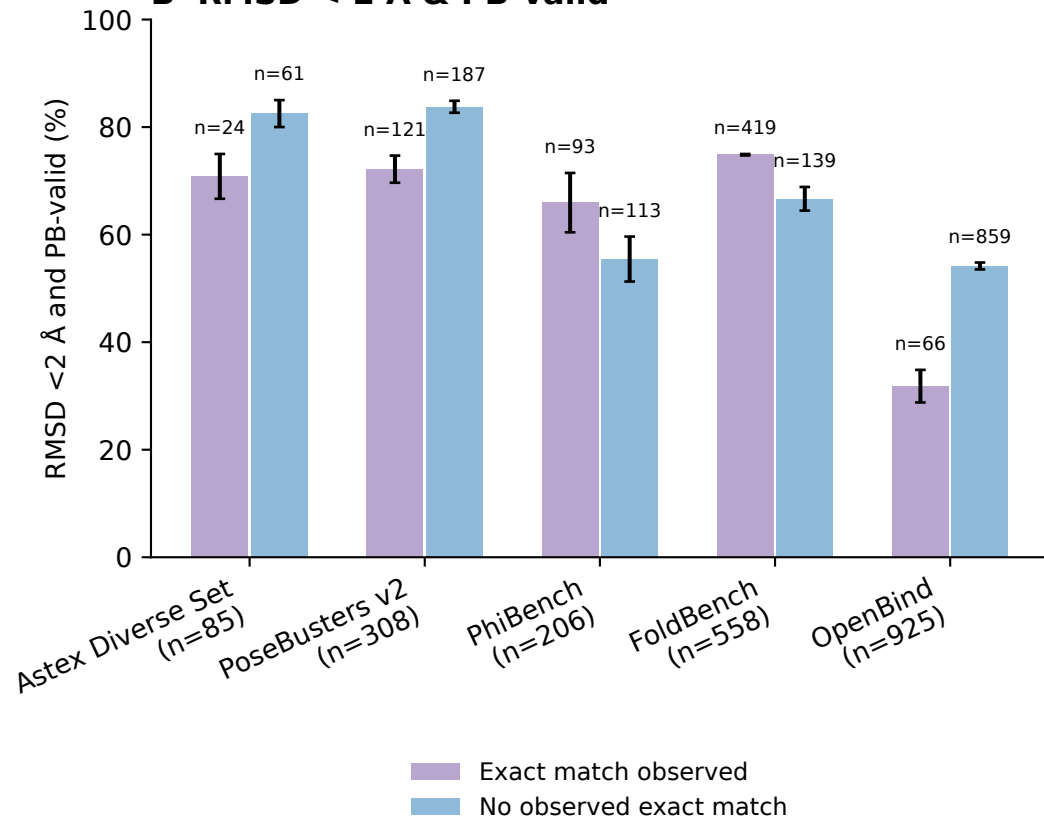
A Sequence identity**B Ligand-protein overlap****C Same training sample**

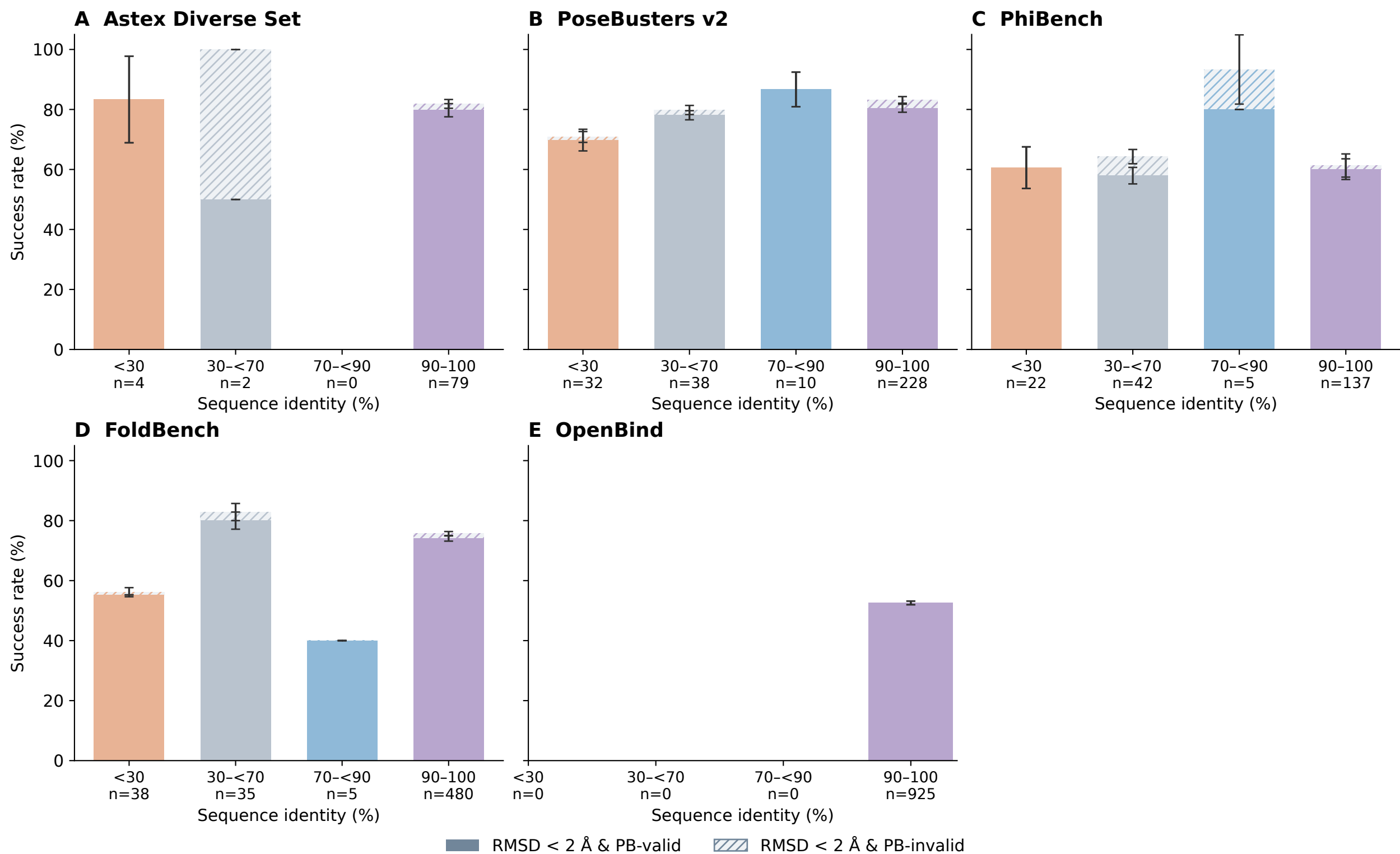
Sequence identity (%)

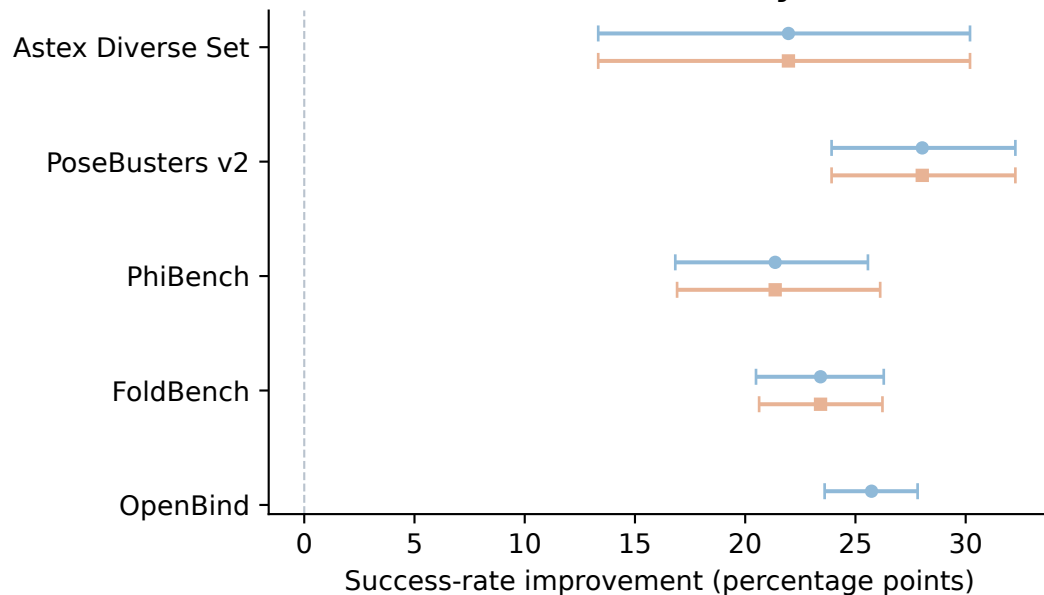
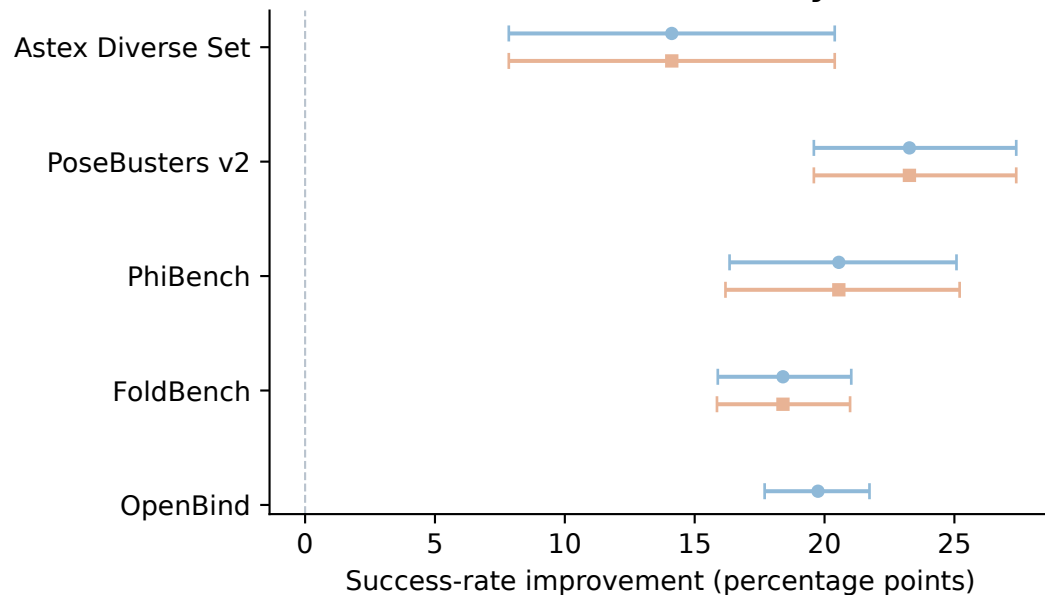
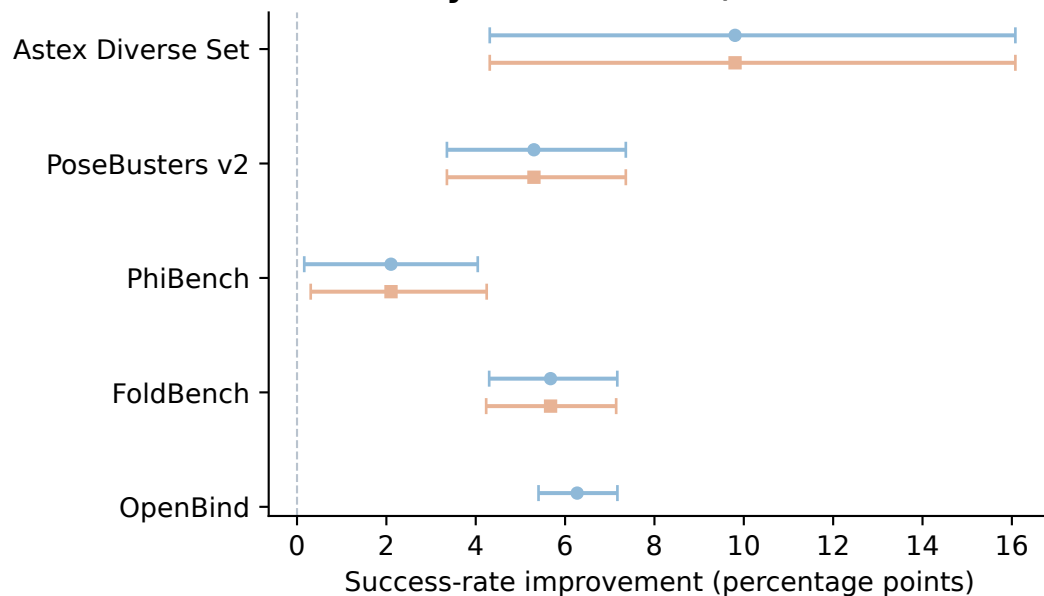
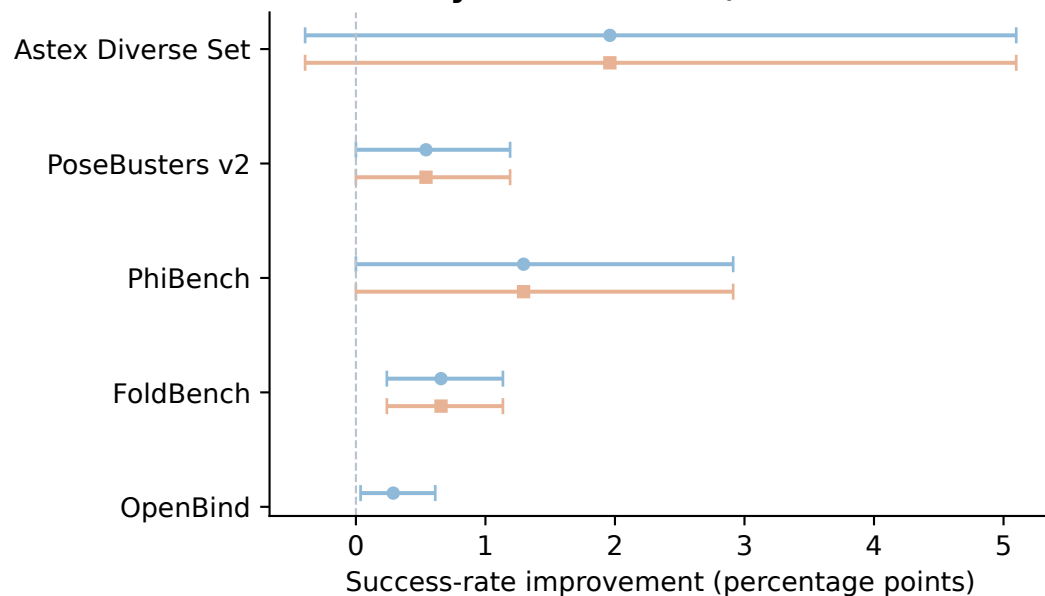
- <30%
- 30-70%
- 70-90%
- 90-100%

Ligand-protein overlap

- Same-sample overlap
- Separate matches
- Ligand only
- Sequence only
- Neither observed

A Ligand similarity**B RMSD < 2 Å & PB-valid**

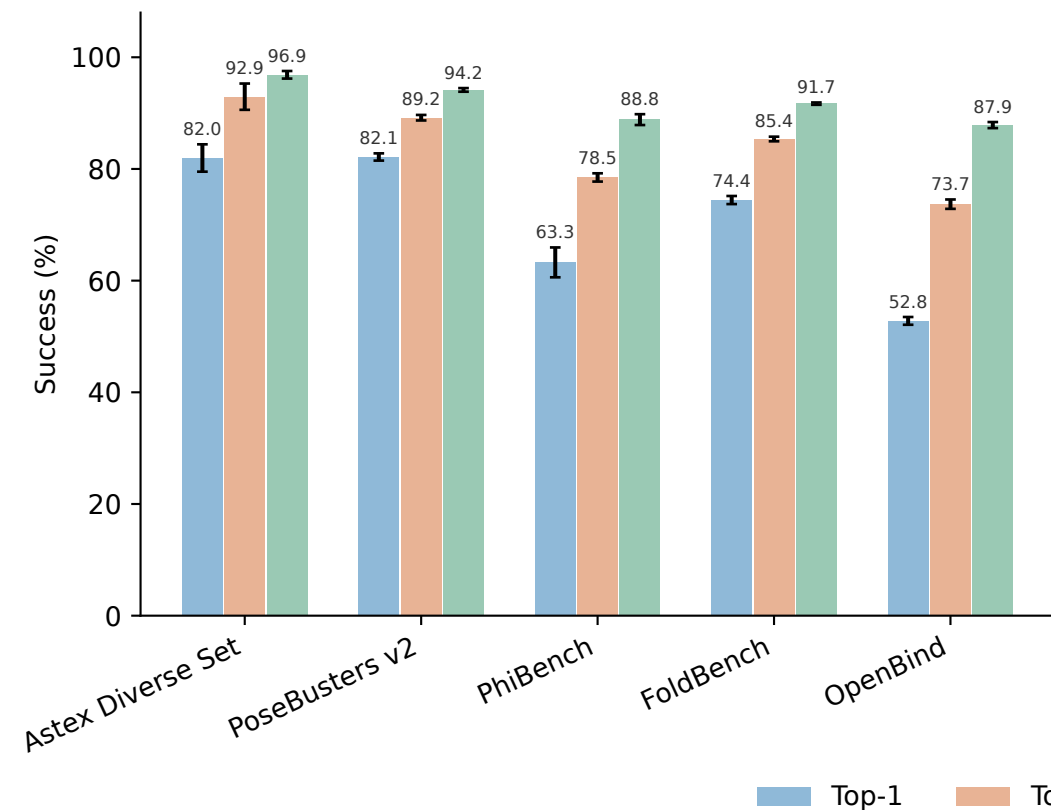
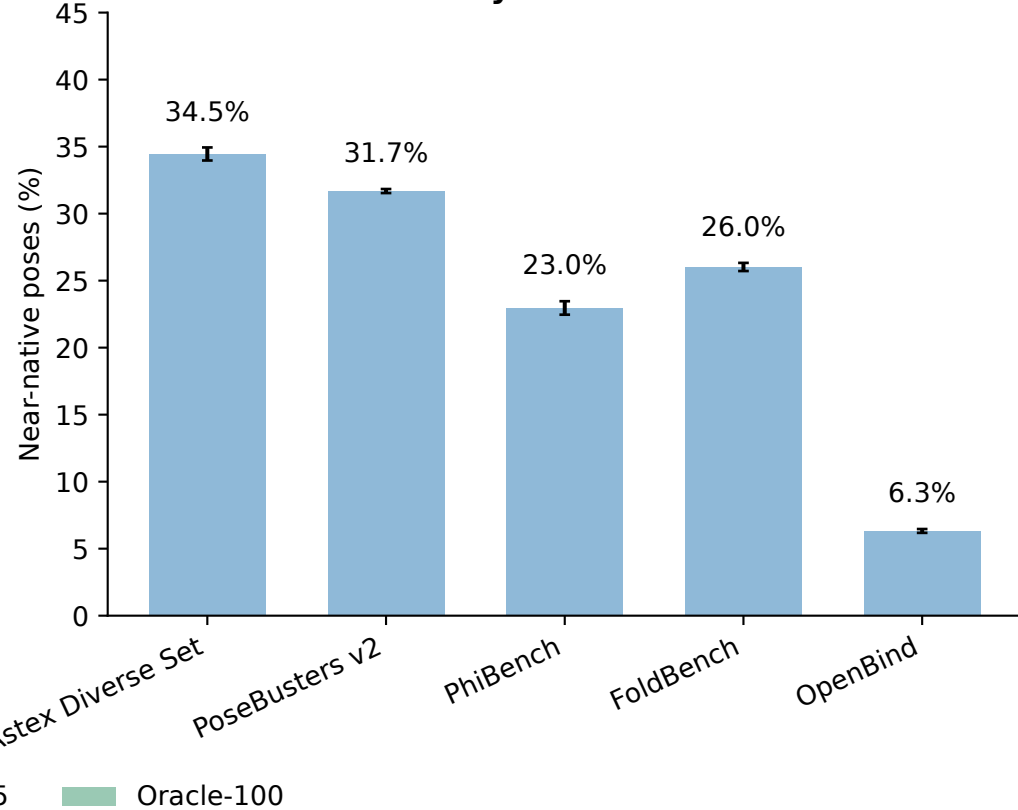


A Refined – raw, no chirality filter**B Refined – raw, with chirality filter****C Chirality filter off → on, raw****D Chirality filter off → on, refined**

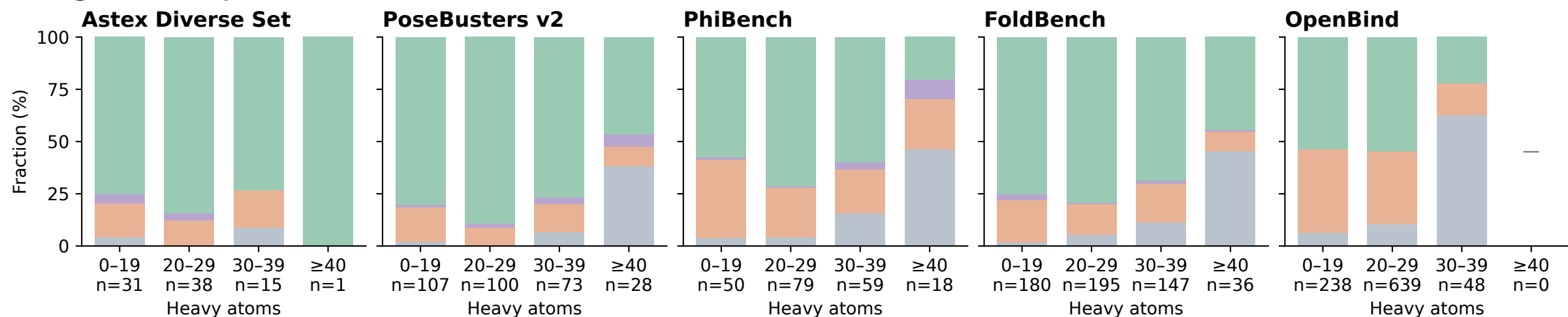
Complex bootstrap



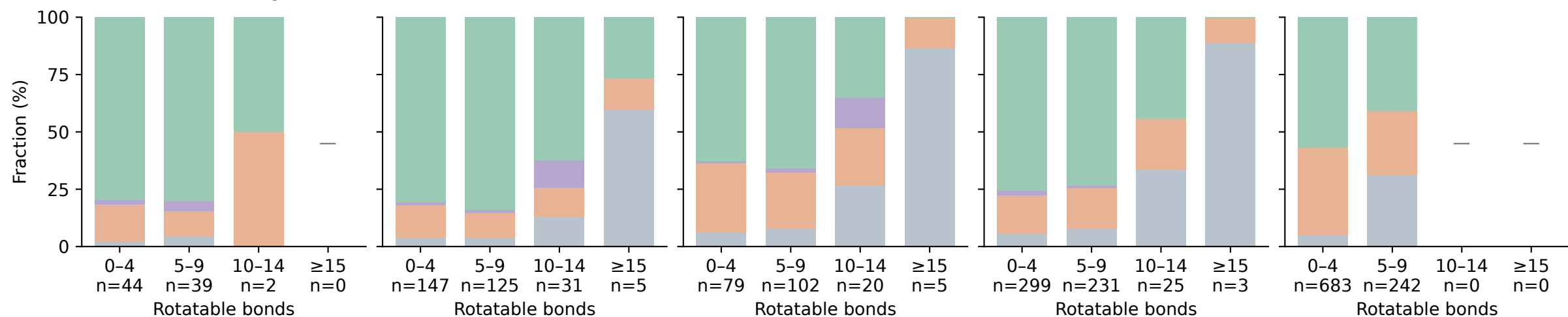
PDB-cluster bootstrap

A Selection**B Near-native density**

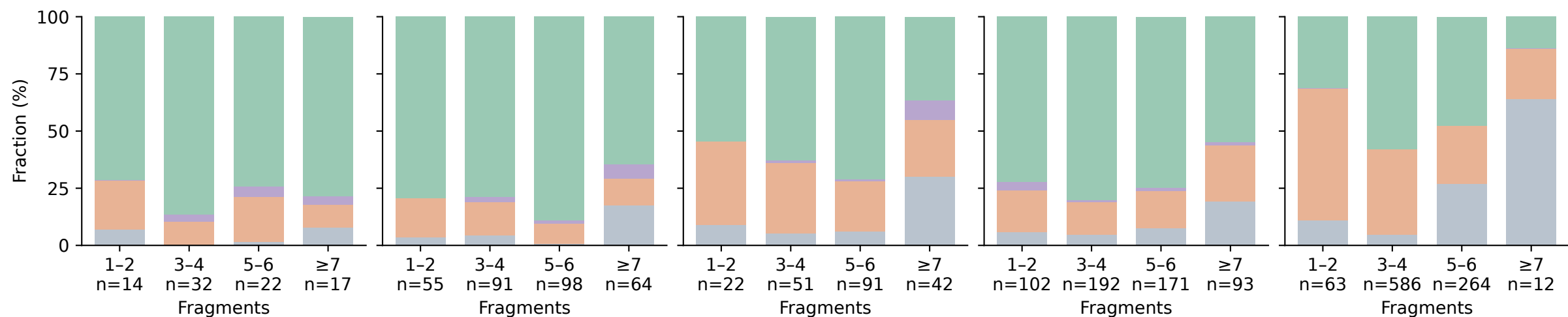
A Ligand size dependence



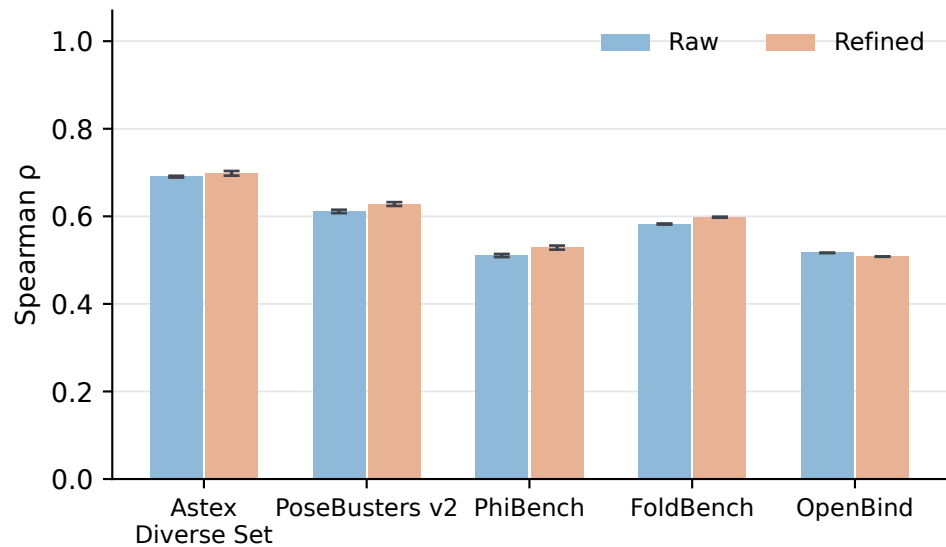
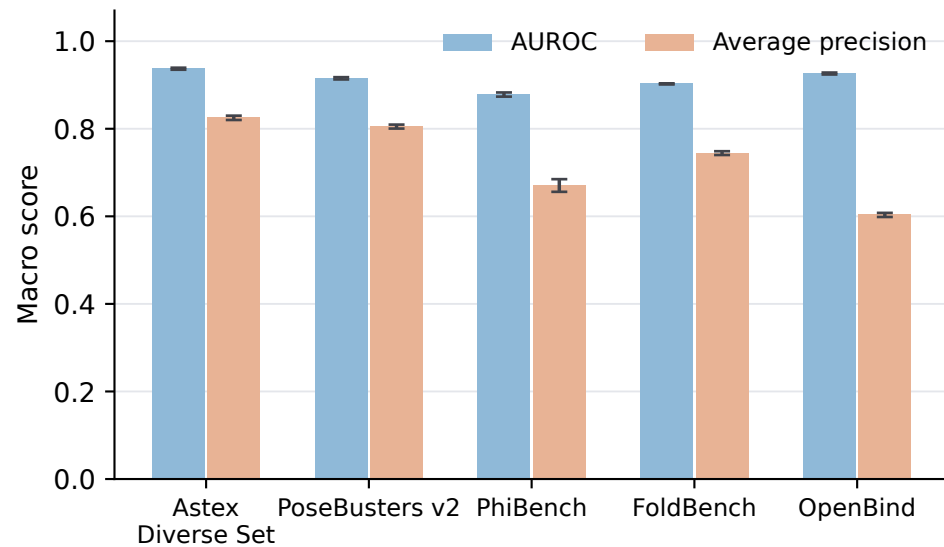
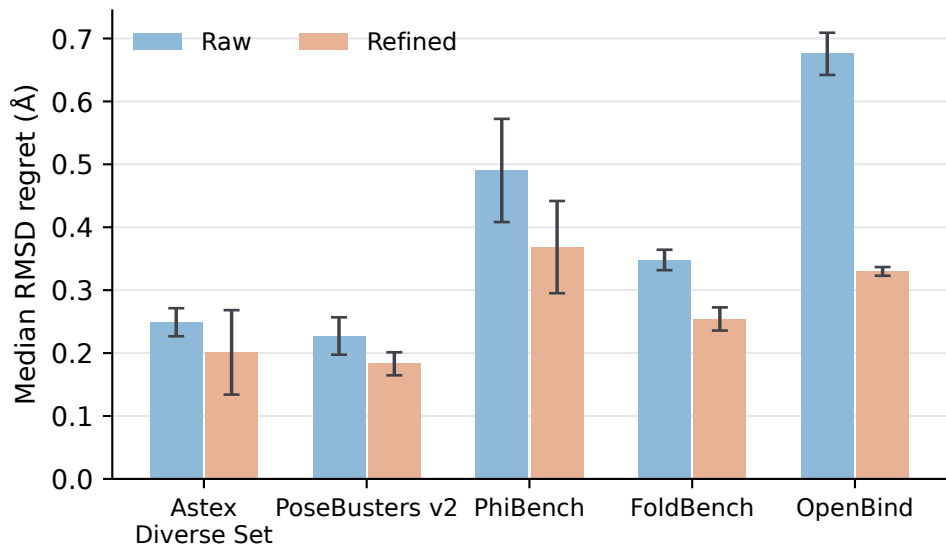
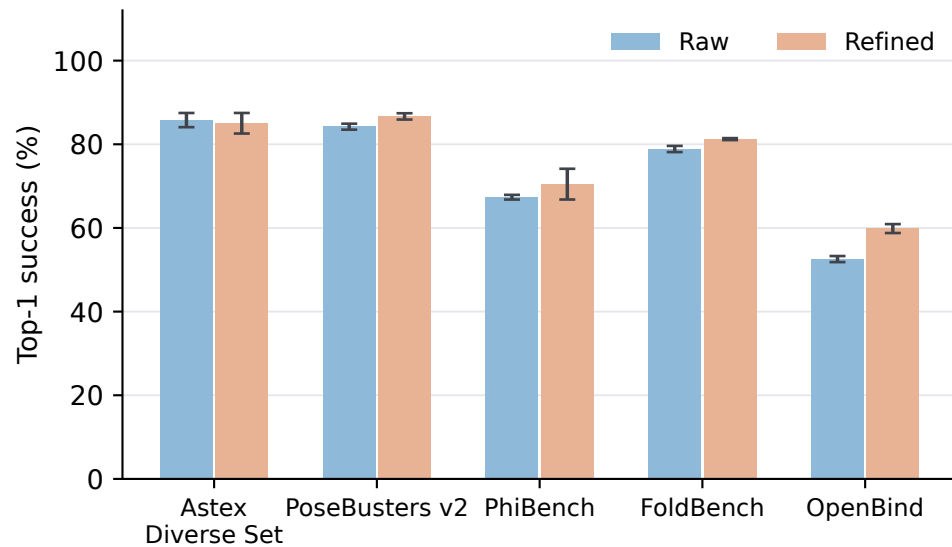
B Rotatable bond dependence

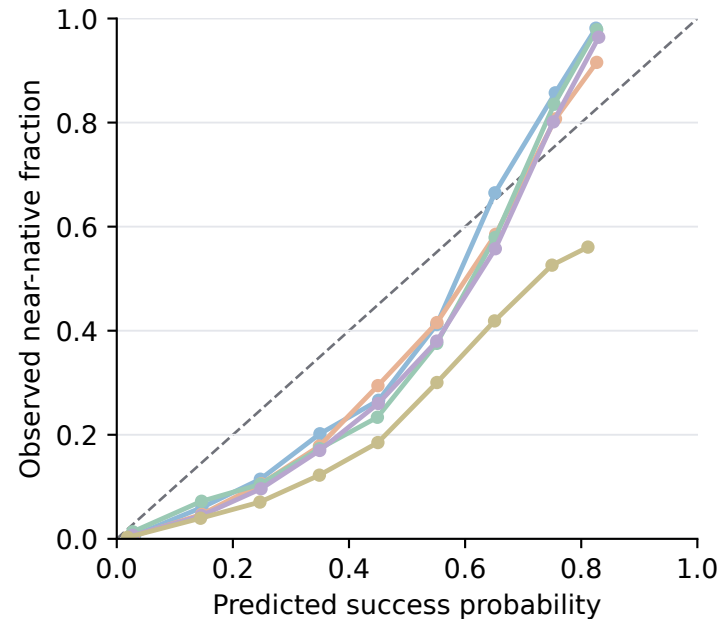
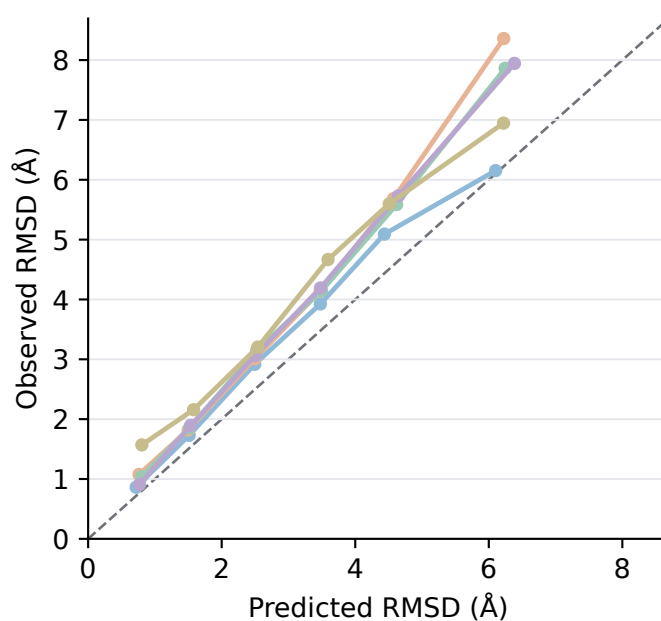
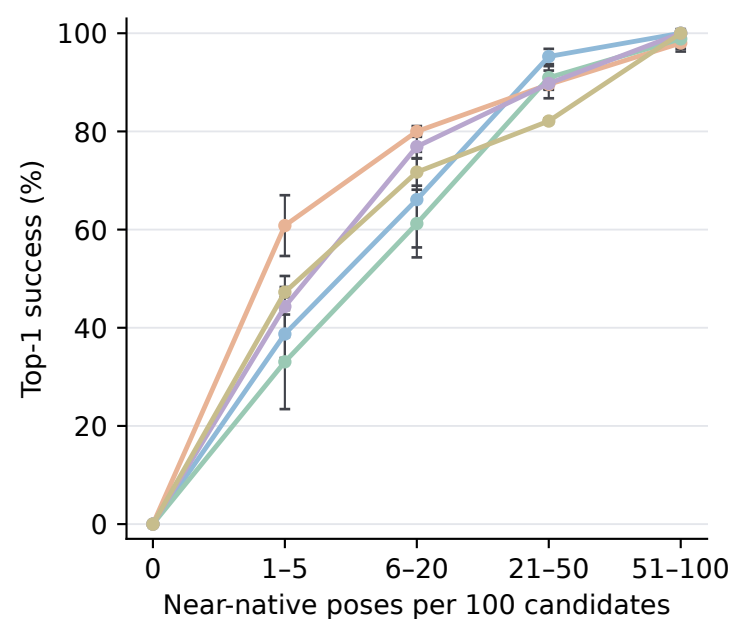


C Fragment count dependence

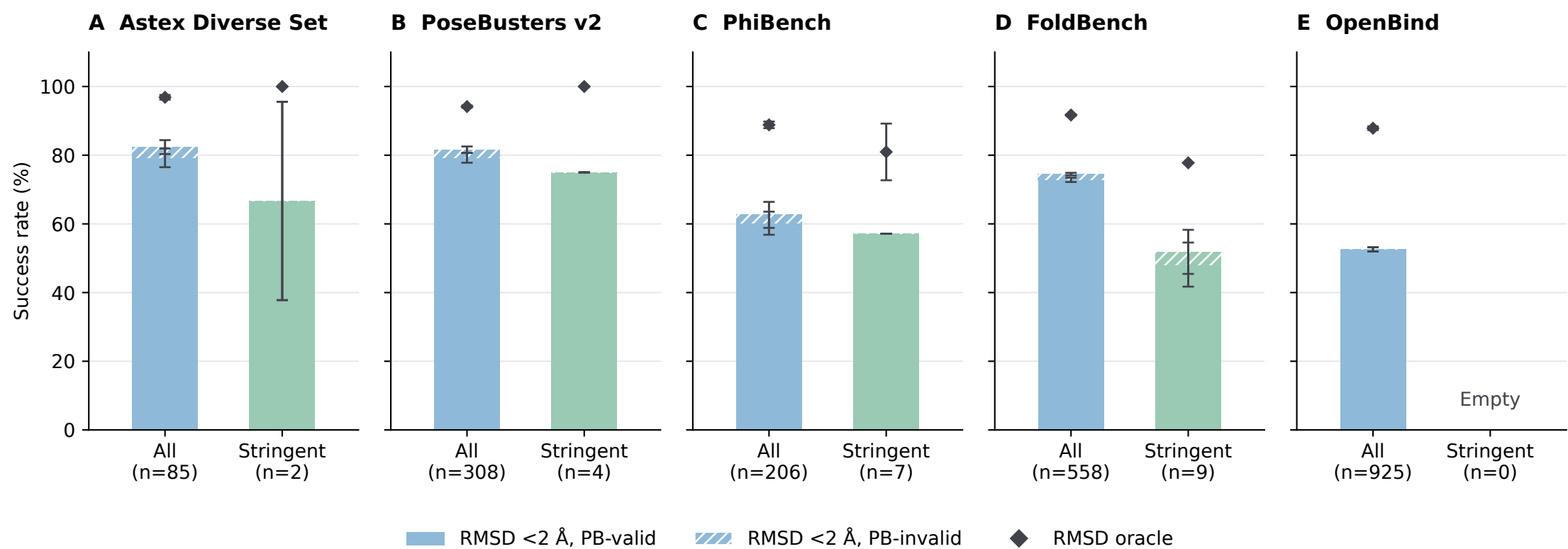


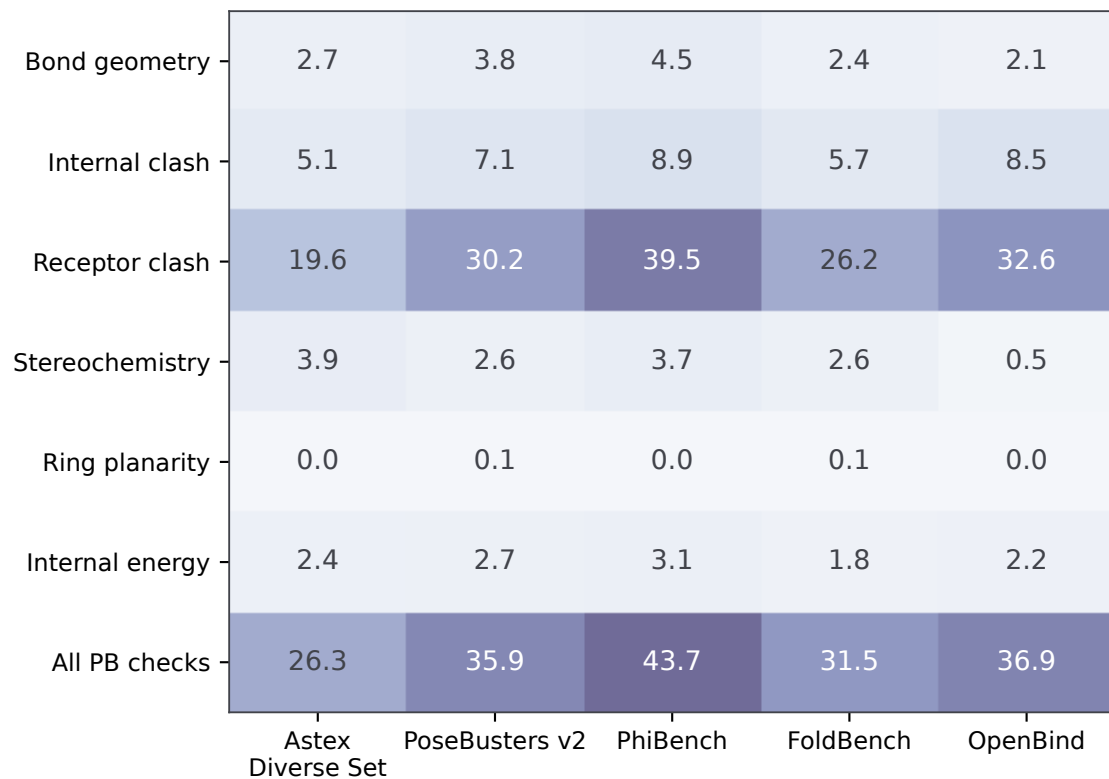
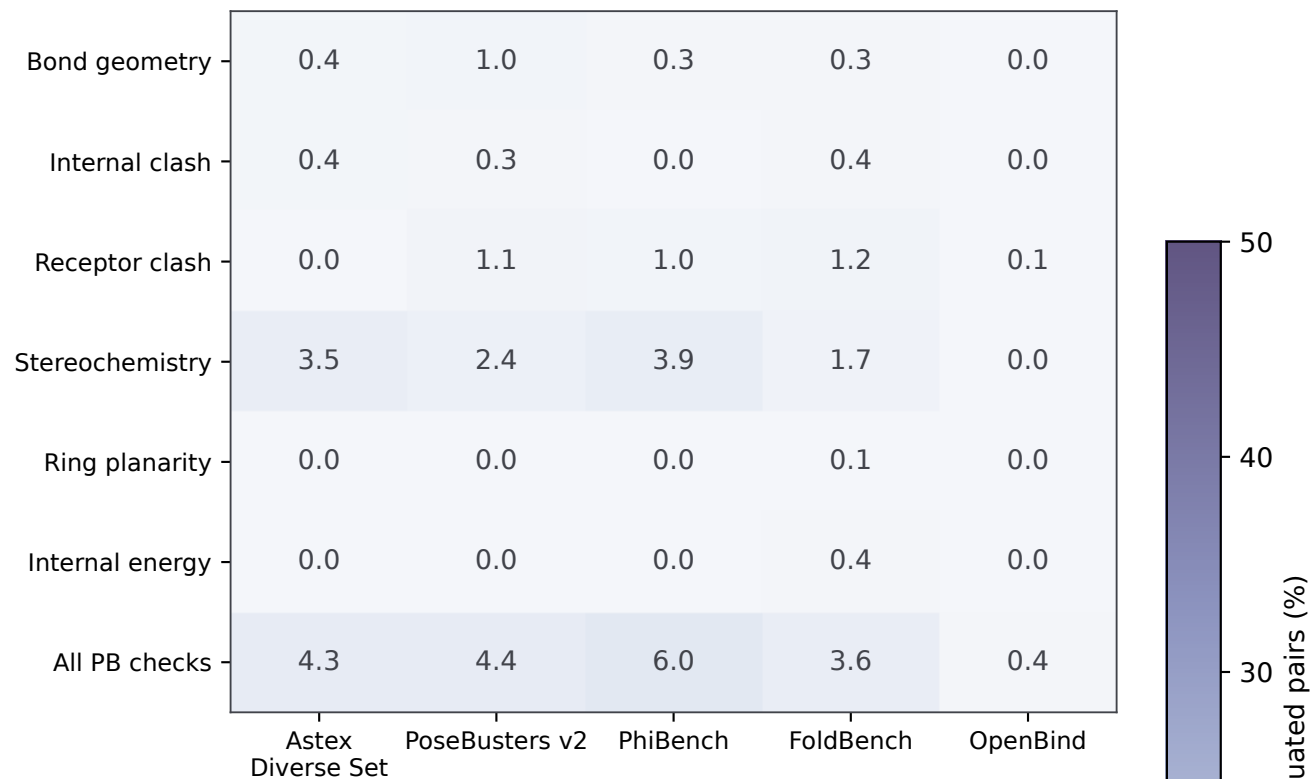
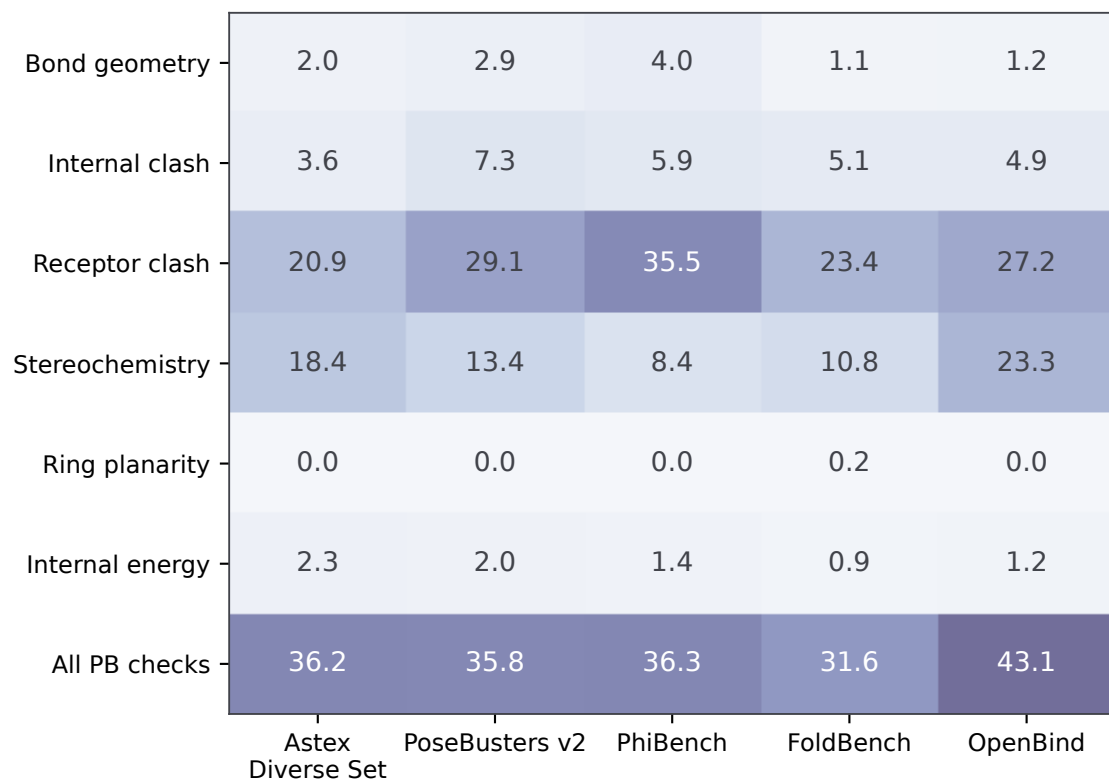
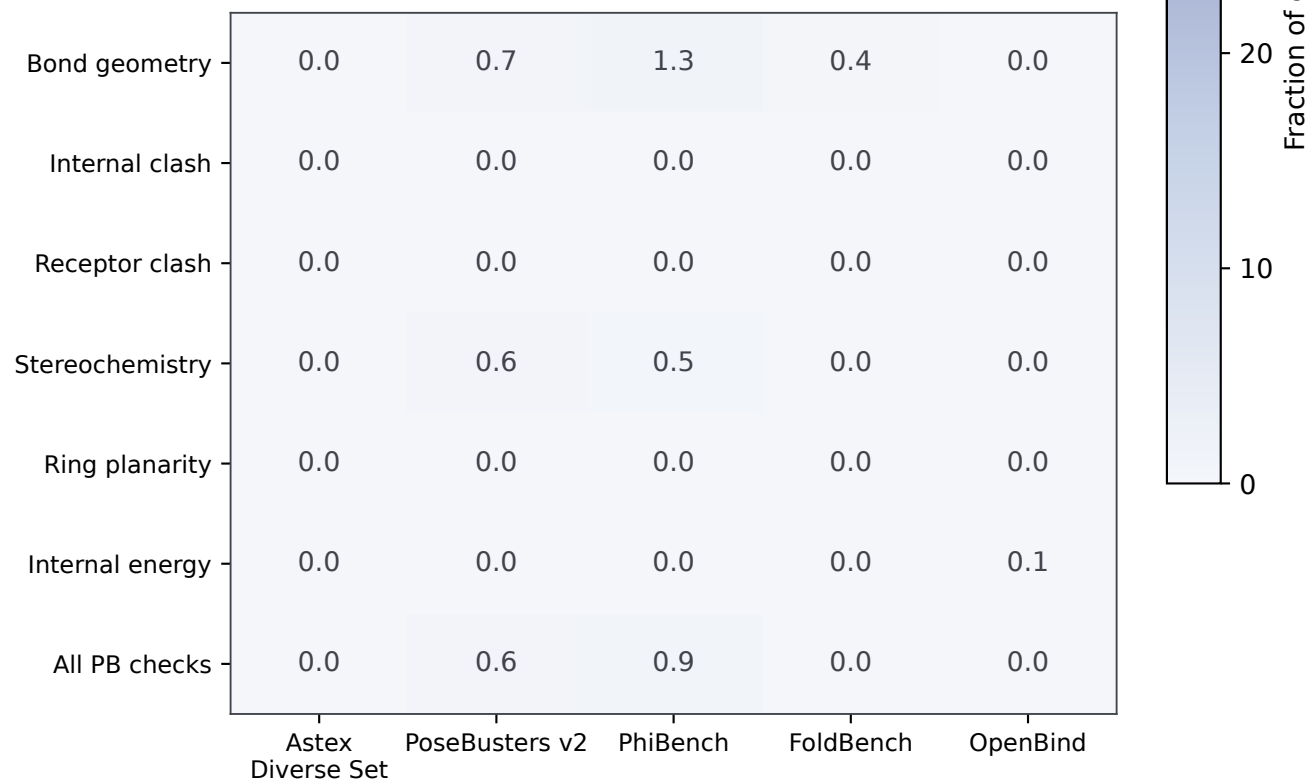
Legend: Near-native pose generation failure (blue), Selection failure (orange), PB failure (purple), Success (green)

A Within-bank ranking**B Near-native discrimination****C Selection regret****D Selection given a successful candidate**

A Success-probability reliability**B RMSD prediction reliability****C Near-native candidate density**

—●— Astex Diverse Set —●— PoseBusters v2 —●— PhiBench —●— FoldBench —●— OpenBind



A Raw selected-pose failure**B Refined selected-pose failure****C Matched candidate: fail → pass****D Matched candidate: pass → fail**

Fraction of evaluated pairs (%)

50

40

30

20

10

0

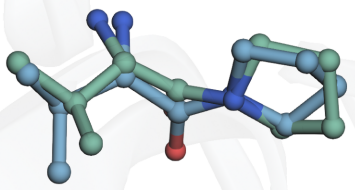
Successful pose

Refinement rescue

Selection failure

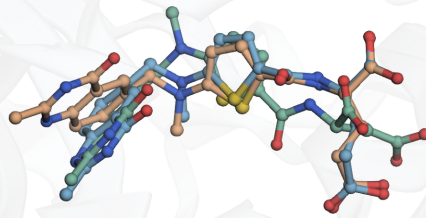
A Astex Diverse Set

1N1M



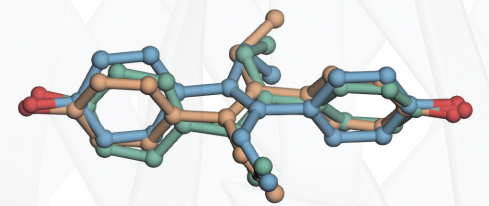
Selected RMSD: 0.75 Å

1HVY



Selected RMSD: 1.98 Å
Raw (same pose): 2.32 Å

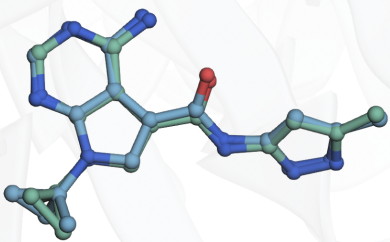
1TZ8



Selected RMSD: 2.27 Å
Oracle RMSD: 0.65 Å

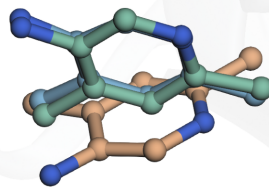
B PoseBusters v2

7DUA · HJ0



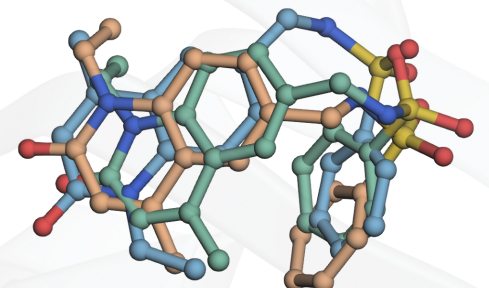
Selected RMSD: 0.88 Å

7FB7 · 8NF



Selected RMSD: 0.30 Å
Raw (same pose): 2.08 Å

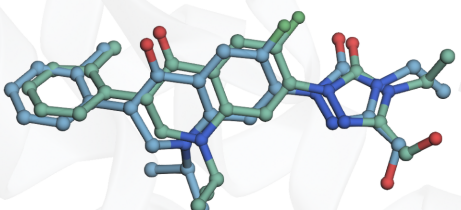
8AY3 · OE3



Selected RMSD: 3.04 Å
Oracle RMSD: 1.17 Å

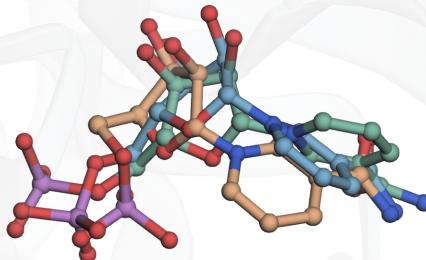
C PhiBench

9DHH · A1A40 · A · 1



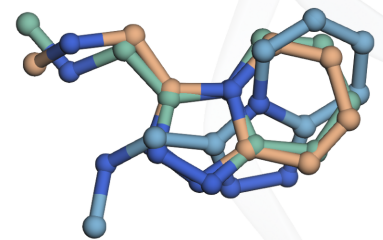
Selected RMSD: 1.05 Å

9FYB · NMN · A · 1 · B



Selected RMSD: 1.51 Å
Raw (same pose): 2.23 Å

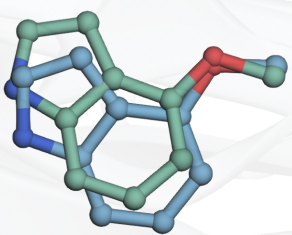
7HHM · UWD · C · 1



Selected RMSD: 2.21 Å
Oracle RMSD: 0.53 Å

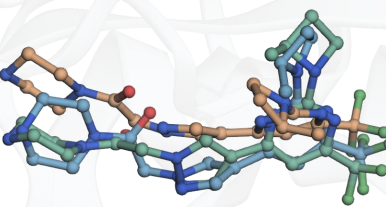
D FoldBench

9BBH · VVP (E)



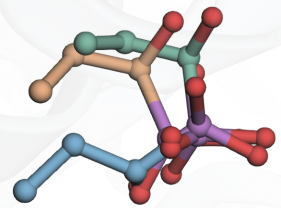
Selected RMSD: 0.91 Å

8UG3 · WRE (C)



Selected RMSD: 1.22 Å
Raw (same pose): 2.52 Å

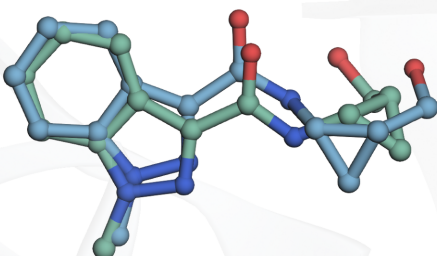
8G7F · 1JJ (H)



Selected RMSD: 2.70 Å
Oracle RMSD: 1.03 Å

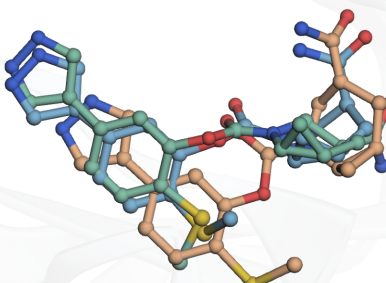
E OpenBind

A71EV2A-X3177A



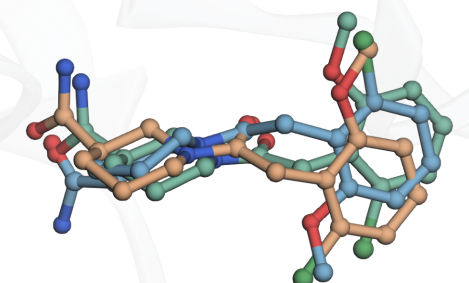
Selected RMSD: 1.12 Å

A71EV2A-X7379A



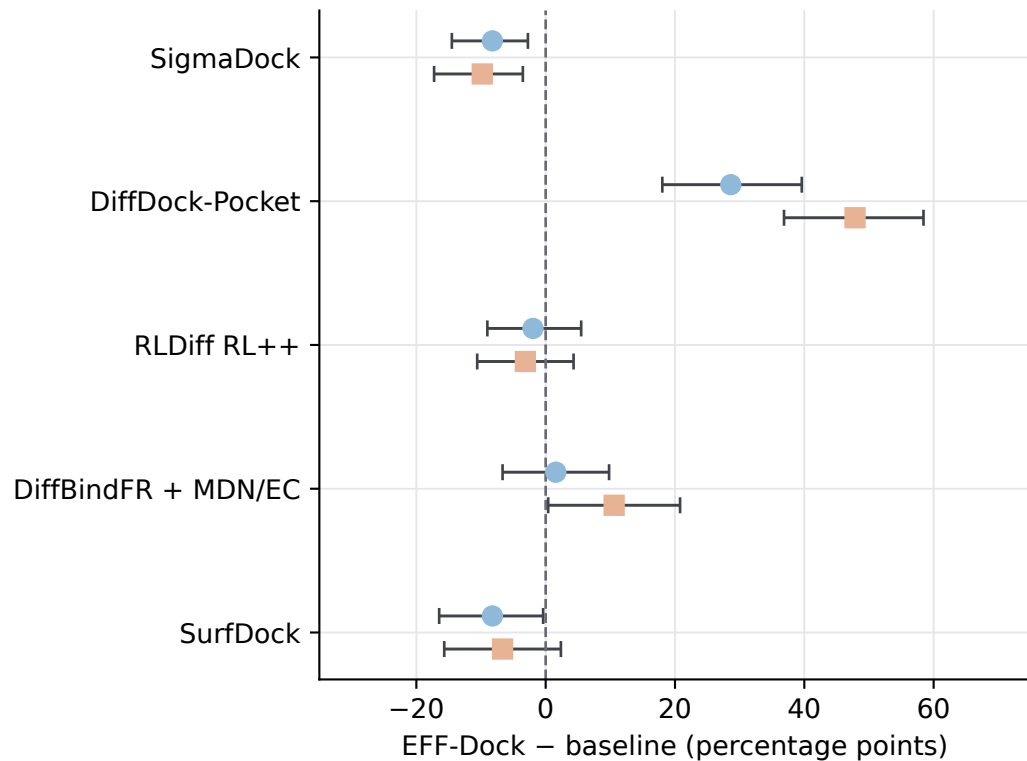
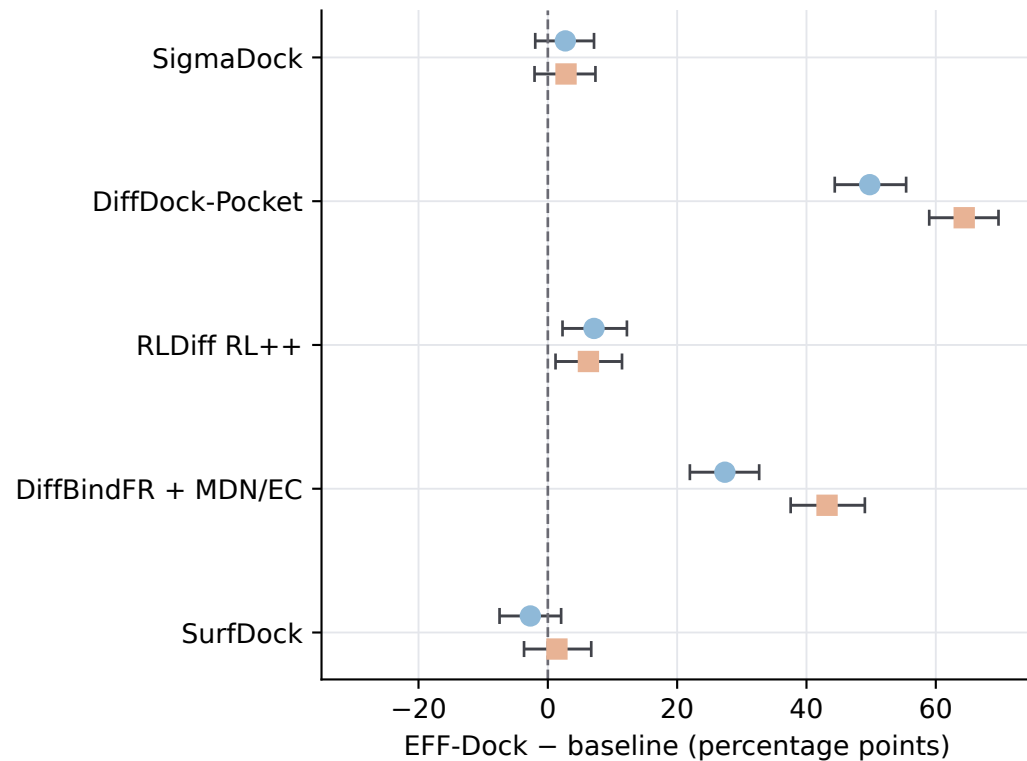
Selected RMSD: 1.60 Å
Raw (same pose): 2.88 Å

A71EV2A-X3858B



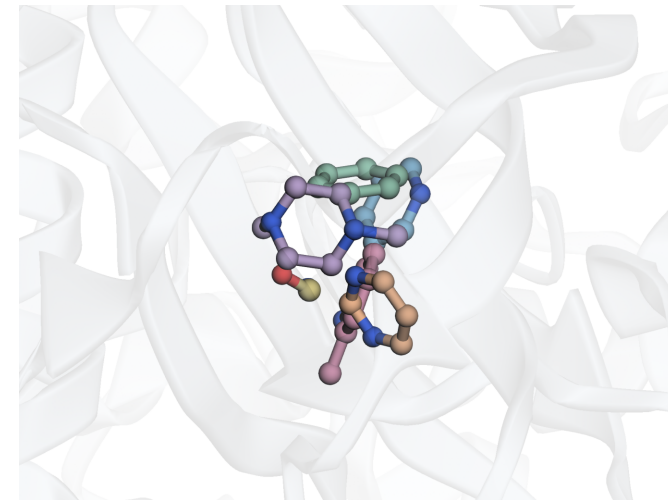
Selected RMSD: 2.64 Å
Oracle RMSD: 1.21 Å

Crystal carbon Selected carbon Raw / oracle carbon N O S P F Cl

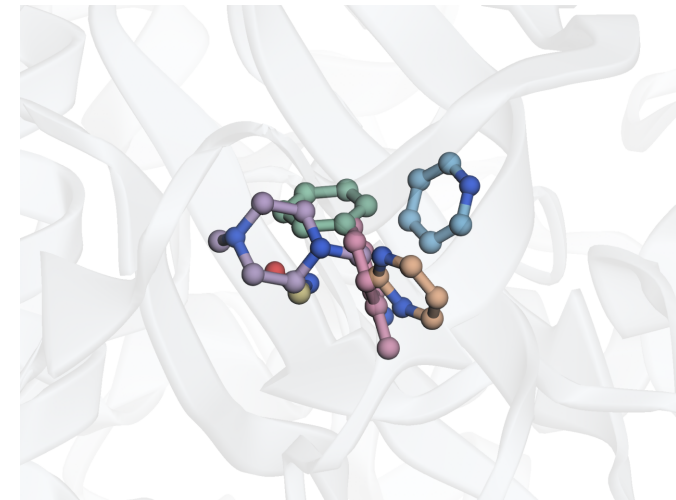
A Astex Diverse Set**B PoseBusters v2**

● RMSD success ■ PB-valid success

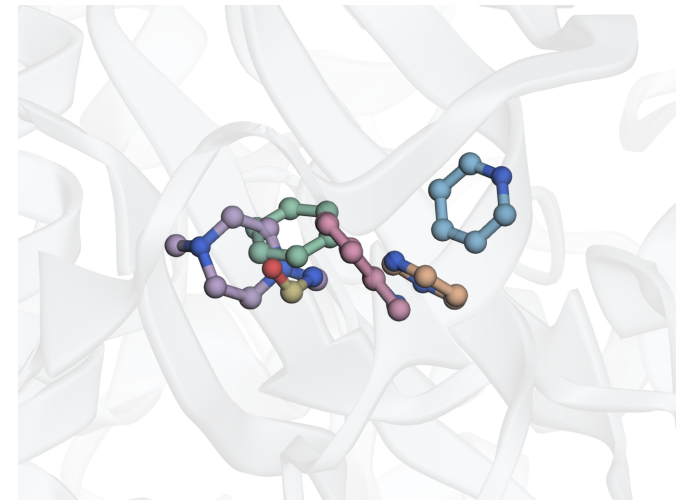
$t = 0.00$



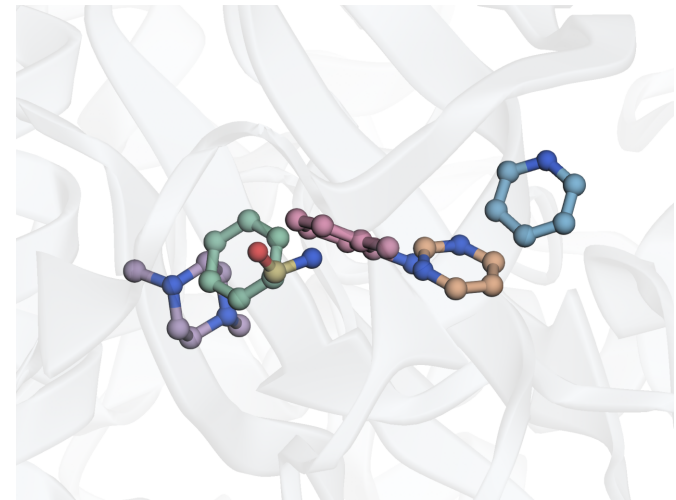
$t = 0.27$



$t = 0.49$



$t = 0.78$



$t = 1.00$

